



ÉCOLE POLYTECHNIQUE FÉDÉRALE DE LAUSANNE

Thermal properties of MOFs from molecular dynamics

MASTER THESIS

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Abstract

Methane-loaded MOF-5 heat capacity is estimated by combining classical NPT molecular dynamics with a loaded-system harmonic quantum correction from PET-JAX/SADMOF Hessians. PET-MAD and PET-SOL calculations at 50, 100, and 150 methane molecules give exploratory room-temperature values of about 0.93, 1.07–1.08, and 1.16 J/g/K. Sampling errors and paired last-layer prediction-rigidity (LLPR) model uncertainties are propagated separately. Production drift, imaginary modes, incomplete relaxation, and sensitivity to uncertainty propagation prevent a claim of convergence. Variation across quenched minima remains unestimated. The empty-framework Hessian provides a separate reference.

1 Scientific motivation and scope

MOF-5 combines high-frequency bond vibrations with soft framework motion; methane adds internal vibrations and anharmonic motion within the pores. Classical MD samples the latter, while normal modes correct the excessive classical occupation of high-frequency vibrations. The conventional host cell contains 424 atoms, $C_{192}H_{96}O_{104}Zn_{32}$; 50, 100, and 150 methane molecules increase the totals to 674, 924, and 1174.

Figure 1 summarizes the calculation. All loaded terms use the same PET checkpoint, composition, and mass normalization. The aim is to test the hybrid approximation of Kapil *et al.* [16] on PET through convergence and validation, rather than matching their force-field results.

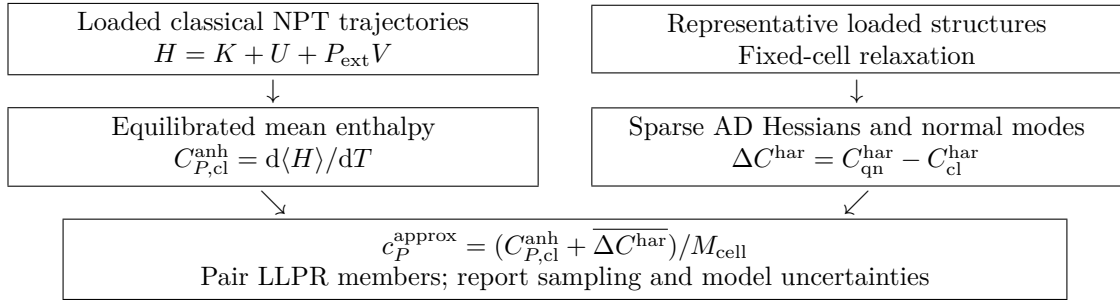


Figure 1: Loaded-system workflow. Accept harmonic references only after force and signed-spectrum checks. The empty-MOF Hessian is evaluated separately, without an empty-MOF MD campaign.

2 Theory

2.1 Born–Oppenheimer approximation: from quantum matter to nuclear motion

2.1.1 The coupled electron–nuclear problem

For electron positions $\mathbf{r}$ and nuclear positions $\mathbf{R}$, the nonrelativistic Hamiltonian at fixed cell $\mathbf{h}$ is

$$\hat{\mathcal{H}}_{\text{tot}} = \hat{K}_n + \hat{K}_e + \hat{V}_{ee} + \hat{V}_{en} + V_{nn}, \quad (1)$$

$$\hat{K}_n = -\sum_{i=1}^N \frac{\hbar^2}{2m_i} \nabla_{\mathbf{R}_i}^2, \quad \hat{K}_e = -\sum_{a=1}^{N_e} \frac{\hbar^2}{2m_e} \nabla_{\mathbf{r}_a}^2. \quad (2)$$

The V terms describe electron–electron, electron–nuclear, and nuclear–nuclear Coulomb interactions. The full wavefunction obeys

$$i\hbar \frac{\partial \Psi(\mathbf{r}, \mathbf{R}, t)}{\partial t} = \hat{\mathcal{H}}_{\text{tot}} \Psi(\mathbf{r}, \mathbf{R}, t). \quad (3)$$

Electron–nuclear attraction couples the two sets of coordinates, making direct propagation impractical for a loaded periodic MOF.

2.1.2 Adiabatic separation and the potential-energy surface

Nuclei are much heavier than electrons. When nuclear motion is slow enough to avoid electronic transitions, solve the electronic ground state at each fixed nuclear geometry [3]:

$$\hat{\mathcal{H}}_{\text{el}}(\mathbf{R}; \mathbf{h}) \psi_0(\mathbf{r}; \mathbf{R}) = \varepsilon_0(\mathbf{R}; \mathbf{h}) \psi_0(\mathbf{r}; \mathbf{R}), \quad (4)$$

$$\hat{\mathcal{H}}_{\text{el}} = \hat{K}_e + \hat{V}_{ee} + \hat{V}_{en}, \quad U(\mathbf{R}; \mathbf{h}) = \varepsilon_0(\mathbf{R}; \mathbf{h}) + V_{nn}(\mathbf{R}; \mathbf{h}). \quad (5)$$

Approximating $\Psi \simeq \psi_0 \chi$ and neglecting nuclear derivatives of ψ_0 removes nonadiabatic couplings and the diagonal adiabatic correction. The remaining nuclear problem is still quantum:

$$\hat{\mathcal{H}}_{\text{BO}} = - \sum_i \frac{\hbar^2}{2m_i} \nabla_{\mathbf{R}_i}^2 + U(\mathbf{R}; \mathbf{h}), \quad \hat{\mathcal{H}}_{\text{BO}} \chi_n = E_n \chi_n. \quad (6)$$

Thus Born–Oppenheimer defines a potential-energy surface, rather than requiring classical nuclei. Here PET approximates that surface without solving the electronic problem at every MD step. Electronic transitions and nonadiabatic effects remain outside both nuclear treatments.

2.1.3 Classical MD and quantum nuclear statistics

Classical MD further replaces nuclear operators by positions and momenta, giving the Hamiltonian in Eq. (15) and Newtonian forces $m_i \dot{\mathbf{R}}_i = -\nabla_i U$. A useful harmonic criterion and thermal length scale are

$$\beta \hbar \omega \ll 1, \quad \lambda_i = \sqrt{\frac{2\pi \hbar^2}{m_i k_B T}}, \quad \beta = (k_B T)^{-1}. \quad (7)$$

Classical sampling is appropriate when quantum level spacing is small relative to thermal energy and nuclear delocalization is negligible on relevant structural scales. A hydrogen-containing system can satisfy these conditions for slow collective modes while violating them for bond vibrations.

Retaining quantum nuclei on the same surface gives the canonical density operator, partition function, and observable average

$$\hat{\rho}_q = Z_q^{-1} e^{-\beta \hat{\mathcal{H}}_{\text{BO}}}, \quad Z_q = \text{Tr} e^{-\beta \hat{\mathcal{H}}_{\text{BO}}} = \sum_n e^{-\beta E_n}, \quad (8)$$

$$\langle A \rangle_q = \text{Tr}(\hat{\rho}_q \hat{A}). \quad (9)$$

The nuclear trace avoids explicit electronic propagation, but not the quantum nuclear problem. PIMD evaluates it through imaginary-time paths: factorizing the exponential and inserting position states produces the ring-polymer distribution in Section 2.9.

We will use the Born–Oppenheimer surface for classical MD and also for PIMD to include nuclear quantum effects, especially in H₂O loadings. Hydrogen’s small mass increases delocalization; O–H zero-point motion and quantum fluctuations of hydrogen bonds motivate this treatment [6]. Their effect on confined-water heat capacity must be evaluated on the selected potential. PIMD retains anharmonic nuclear sampling, rather than adding a harmonic correction to classical motion. The methane results reported here still use classical MD with the separate harmonic correction; water PIMD is a planned extension.

2.2 Density functional theory and the MLIP reference data

Density functional theory (DFT) is a quantum-mechanical method for computing the electronic structure and energy of an atomic configuration. Within the Born–Oppenheimer approximation, the nuclei are held fixed while the electronic ground state is determined. Its central variable is the electron density $n(\mathbf{r})$, the number of electrons per unit volume, rather than the full many-electron wavefunction. The Hohenberg–Kohn theorems establish that the ground-state energy can be expressed as a functional of this density: a rule assigning an energy to an entire density distribution. Minimizing the exact functional at fixed electron number gives the ground-state density and energy [4]. In the Kohn–Sham decomposition, the electronic energy functional is

$$E_{\text{el}}[n; \mathbf{R}, \mathbf{h}] = T_s[n] + \int_{\text{cell}} v_{\text{ext}}(\mathbf{r}; \mathbf{R}, \mathbf{h}) n(\mathbf{r}) d\mathbf{r} + E_{\text{H}}[n] + E_{\text{xc}}[n], \quad (10)$$

where T_s is the kinetic energy of the auxiliary noninteracting electrons, v_{ext} is the nuclear potential, E_{H} is the classical electron–electron electrostatic energy, and E_{xc} is the exchange–correlation energy. The ground-state density minimizes this functional subject to $\int_{\text{cell}} n(\mathbf{r}) d\mathbf{r} = N_e$.

In the usual Kohn–Sham formulation, an auxiliary system of noninteracting electrons reproduces the density of the interacting system [5]. Its one-electron equations contain an effective potential combining nuclear attraction, the electrostatic field of the electron density, and exchange–correlation effects:

$$\left[-\frac{\hbar^2}{2m_e} \nabla^2 + v_{\text{eff}}[n](\mathbf{r}) \right] \phi_a(\mathbf{r}) = \epsilon_a \phi_a(\mathbf{r}), \quad (11)$$

$$v_{\text{eff}}[n](\mathbf{r}) = v_{\text{ext}}(\mathbf{r}) + \frac{\delta E_{\text{H}}}{\delta n(\mathbf{r})} + \frac{\delta E_{\text{xc}}}{\delta n(\mathbf{r})}, \quad (12)$$

$$n(\mathbf{r}) = \sum_a f_a |\phi_a(\mathbf{r})|^2, \quad \sum_a f_a = N_e. \quad (13)$$

Here ϕ_a and ϵ_a are the auxiliary orbitals and their eigenvalues, and f_a are their occupation numbers; for a periodic solid, the sum includes band and wavevector weights. The exchange–correlation functional accounts for the remaining many-electron contributions and must be approximated in practical calculations. Starting from a trial density, the equations are solved to obtain electronic orbitals and an updated density; this cycle is repeated until the density and energy are self-consistent: the output density must reproduce the density used to build v_{eff} . At the converged density n_0 , adding nuclear repulsion gives the potential energy and its nuclear derivatives give the reference forces:

$$U_{\text{DFT}}(\mathbf{R}; \mathbf{h}) = E_{\text{el}}[n_0; \mathbf{R}, \mathbf{h}] + V_{\text{nn}}(\mathbf{R}; \mathbf{h}), \quad \mathbf{F}_i^{\text{DFT}} = -\nabla_{\mathbf{R}_i} U_{\text{DFT}}|_{\mathbf{h}}. \quad (14)$$

Derivatives with respect to cell strain similarly provide reference stresses.

These DFT energies, forces, and stresses provide reference labels for training a machine-learned interatomic potential (MLIP). PET-MAD, for example, is pretrained on a chemically diverse DFT dataset [18]. The MLIP learns to predict the reference potential-energy surface directly from atomic configurations, making long MD trajectories and Hessian calculations affordable without repeating the electronic calculation at each step. Its accuracy depends both on the DFT approximations used to generate the labels and on how well the training configurations cover the environments encountered in the simulation.

2.3 Statistical mechanics for molecular dynamics

The required observable is the fixed-loading, constant-pressure heat capacity. Equilibrium distributions define its averages; correlations limit the information in finite trajectories [1, 2].

2.3.1 Microscopic model and equations of motion

Taking the classical nuclear limit of Section 2.1, with momenta $\mathbf{p}_i = m_i \mathbf{v}_i$ and volume $V = \det \mathbf{h}$, gives

$$\mathcal{H}(\mathbf{R}, \mathbf{p}; \mathbf{h}) = K(\mathbf{p}) + U(\mathbf{R}; \mathbf{h}), \quad K = \sum_{i=1}^N \frac{\mathbf{p}_i^2}{2m_i}. \quad (15)$$

Physical internal energy is $E = K + U$, and enthalpy is $H = E + P_{\text{ext}}V$. Hamilton’s equations give the forces:

$$\dot{\mathbf{R}}_i = \frac{\partial \mathcal{H}}{\partial \mathbf{p}_i} = \frac{\mathbf{p}_i}{m_i}, \quad \dot{\mathbf{p}}_i = -\frac{\partial \mathcal{H}}{\partial \mathbf{R}_i} = \mathbf{F}_i. \quad (16)$$

Exact fixed-cell dynamics conserves E for a time-independent potential. Reservoir couplings exchange heat and volume. Classical nuclei omit zero-point motion, tunneling, and quantum populations; a vibrational mode is approximately classical when $k_{\text{B}}T \gg \hbar\omega$.

2.3.2 Phase space and equilibrium averages

A microstate is $\Gamma = (\mathbf{R}, \mathbf{p})$. For unconstrained Cartesian nuclei,

$$d\Gamma = \frac{d^{3N}R d^{3N}p}{h_{\text{P}}^{3N} \prod_s N_s!}, \quad (17)$$

Here h_{P} is Planck’s constant and N_s counts identical particles of species s . These normalization factors cancel in fixed-composition averages; constraints require a reduced measure.

$$\langle A \rangle = \int d\Gamma \rho(\Gamma) A(\Gamma), \quad \int d\Gamma \rho(\Gamma) = 1. \quad (18)$$

Hamiltonian flow preserves phase-space volume:

$$\frac{\partial \rho}{\partial t} + \sum_i \left(\dot{\mathbf{R}}_i \cdot \nabla_{\mathbf{R}_i} \rho + \dot{\mathbf{p}}_i \cdot \nabla_{\mathbf{p}_i} \rho \right) = 0. \quad (19)$$

Equating an ensemble average with a long-time average additionally assumes ergodicity:

$$\langle A \rangle = \lim_{\tau \rightarrow \infty} \frac{1}{\tau} \int_0^\tau A(\Gamma(t)) dt. \quad (20)$$

Equilibration, barrier crossing, and guest redistribution must be tested with structural diagnostics and independent replicas; stationary temperature alone does not establish adequate sampling.

2.3.3 Equilibrium ensembles

NVE fixes energy and volume; NVT exchanges heat; NPT also exchanges volume. All retain fixed composition and therefore do not determine an adsorption isotherm. Their equilibrium weights are

$$\rho_{NVE}(\Gamma) = \frac{\delta(E_0 - \mathcal{H}(\Gamma))}{\Omega(N, V, E_0)}, \quad (21)$$

where $\Omega = \int d\Gamma \delta(E_0 - \mathcal{H})$ and additional invariants restrict the energy shell. NVE is useful for energy-conservation tests. With $\beta = (k_B T)^{-1}$, canonical sampling uses

$$Z(N, V, T) = \int d\Gamma e^{-\beta \mathcal{H}(\Gamma)}, \quad (22)$$

$$\rho_{NVT}(\Gamma) = Z^{-1} e^{-\beta \mathcal{H}(\Gamma)}. \quad (23)$$

Integrating momenta gives a configurational weight $e^{-\beta U}$. Partition functions connect these distributions to thermodynamics:

$$F = -k_B T \ln Z, \quad \langle E \rangle = - \left(\frac{\partial \ln Z}{\partial \beta} \right)_{N, V}, \quad (24)$$

$$S = - \left(\frac{\partial F}{\partial T} \right)_{N, V}, \quad P = - \left(\frac{\partial F}{\partial V} \right)_{N, T}. \quad (25)$$

For variable volume at fixed cell shape,

$$\Delta(N, P_{\text{ext}}, T) = \int_0^\infty \frac{dV}{V_0} e^{-\beta P_{\text{ext}} V} Z(N, V, T), \quad (26)$$

where V_0 is a fixed reference volume, and

$$\rho_{NPT}(\Gamma, V) = \Delta^{-1} e^{-\beta[\mathcal{H}(\Gamma; V) + P_{\text{ext}} V]}. \quad (27)$$

Thus $G = -k_B T \ln \Delta$, $\langle H \rangle = -\partial_\beta \ln \Delta$, and $\langle V \rangle = (\partial G / \partial P_{\text{ext}})_{N, T}$. Use imposed pressure in H ; exclude thermostat and barostat energies.

Fractional coordinates $\mathbf{s}_i = \mathbf{h}^{-1} \mathbf{R}_i$ give $d^{3N} R = V^N d^{3N} s$. Flexible-cell NPT needs the correct cell measure and extended-system corrections [21]; the present calculation permits anisotropic strain under hydrostatic external pressure.

2.3.4 Equipartition and temperature

Each independent quadratic term contributes $k_B T/2$. For f active momentum degrees of freedom,

$$\langle K \rangle = \frac{f}{2} k_B T, \quad T_{\text{kin}} = \frac{2K}{f k_B}. \quad (28)$$

Subtract constraints and deliberately removed momentum components from $3N$; fixing total linear momentum gives $f = 3N - 3$. Auxiliary reservoir variables do not count. Initial velocities follow

$$v_{i\alpha} \sim \mathcal{N}\left(0, \frac{k_B T}{m_i}\right), \quad \alpha \in \{x, y, z\}. \quad (29)$$

Subtract center-of-mass velocity when required. Canonical fluctuations obey $\text{var}(K) = f(k_B T)^2/2$ and $\text{var}(T_{\text{kin}}) = 2T^2/f$; repeated exact rescaling suppresses these fluctuations.

2.3.5 Response functions and heat capacity

For a temperature-independent observable and Hamiltonian,

$$\left(\frac{\partial \langle A \rangle}{\partial T} \right)_{N, V} = \frac{\text{cov}(A, E)}{k_B T^2}, \quad (30)$$

where $\text{cov}(A, B) = \langle AB \rangle - \langle A \rangle \langle B \rangle$. At fixed pressure, replace E by physical enthalpy H :

$$C_V = \left(\frac{\partial \langle E \rangle}{\partial T} \right)_{N, V} = \frac{\text{var}_{NVT}(E)}{k_B T^2}, \quad (31)$$

$$C_P = \left(\frac{\partial \langle H \rangle}{\partial T} \right)_{N, P_{\text{ext}}} = \frac{\text{var}_{NPT}(H)}{k_B T^2}. \quad (32)$$

For separable kinetic energy, equivalent configurational estimators are

$$C_V = \frac{f}{2}k_B + \frac{\text{var}_{NVT}(U)}{k_B T^2}, \quad (33)$$

$$C_P = \frac{f}{2}k_B + \frac{\text{var}_{NPT}(U + P_{\text{ext}}V)}{k_B T^2}. \quad (34)$$

Omitting the kinetic term underestimates heat capacity. The NVT variance formula cannot be applied to constant-energy NVE trajectories. Volume responses similarly satisfy

$$\kappa_T = -\frac{1}{\langle V \rangle} \left(\frac{\partial \langle V \rangle}{\partial P_{\text{ext}}} \right)_{N,T} = \frac{\text{var}_{NPT}(V)}{k_B T \langle V \rangle}, \quad (35)$$

$$\alpha_V = \frac{1}{\langle V \rangle} \left(\frac{\partial \langle V \rangle}{\partial T} \right)_{N,P_{\text{ext}}} = \frac{\text{cov}_{NPT}(V, H)}{k_B T^2 \langle V \rangle}. \quad (36)$$

For a stable hydrostatic phase,

$$C_P - C_V = \frac{TV\alpha_V^2}{\kappa_T}, \quad (37)$$

Explicitly state-dependent effective potentials require additional response terms. The present workflow instead differentiates mean NPT enthalpies across temperatures (Eq. (98)). Smaller temperature spacing reduces discretization bias but amplifies noise. Cell heat capacities become mass-specific values through $c_P = C_P/M_{\text{cell}}$, including guest mass.

2.3.6 Correlation and sampling uncertainty

Discard initial transients using running/block averages, structure, and replica comparisons. Minimization does not replace thermal equilibration. For M production samples saved every δt_s ,

$$\bar{A} = \frac{1}{M} \sum_{n=1}^M A_n, \quad (38)$$

$$C_A(k) = \langle (A_n - \langle A \rangle)(A_{n+k} - \langle A \rangle) \rangle, \quad (39)$$

$$\rho_A(k) = C_A(k)/C_A(0). \quad (40)$$

The sample-mean variance is

$$\text{var}(\bar{A}) = \frac{C_A(0)}{M} \left[1 + 2 \sum_{k=1}^{M-1} \left(1 - \frac{k}{M} \right) \rho_A(k) \right]. \quad (41)$$

For runs much longer than the correlation time,

$$g_A = 1 + 2 \sum_{k=1}^{\infty} \rho_A(k), \quad \tau_{\text{int},A} = \frac{\delta t_s}{2} g_A, \quad (42)$$

$$M_{\text{eff},A} \simeq \frac{M}{g_A}, \quad \text{SE}(\bar{A}) \simeq \sqrt{\frac{C_A(0)}{M_{\text{eff},A}}}. \quad (43)$$

Estimate or truncate the correlation sum statistically. Effective sample counts depend on the observable; finer output stride does not add independent information over a fixed duration.

Block errors should plateau for blocks longer than the correlation time, while retaining enough blocks [8]. Independent replicas help detect slow modes, but neither method reveals a basin never visited. Resample whole replicas or sufficiently long blocks and recompute the full estimator, preserving cross-temperature correlations. Variance-based heat capacities require errors of squared fluctuations, not merely errors of the mean enthalpy. Sampling error, finite-size effects, integration bias, and potential error need separate checks.

2.3.7 Configurational populations

For temperature-independent basins $\mathcal{B}_a$,

$$\Delta = \sum_a \Delta_a, \quad w_a = \frac{\Delta_a}{\Delta}, \quad \langle H \rangle = \sum_a w_a \langle H \rangle_a. \quad (44)$$

$$C_P = \sum_a w_a \frac{\partial \langle H \rangle_a}{\partial T} + \sum_a \langle H \rangle_a \frac{\partial w_a}{\partial T}. \quad (45)$$

The two terms describe motion within basins and redistribution between them. A Hessian characterizes one local environment; adequately sampled MD also captures changing guest arrangements and framework conformations. Basin weights depend on accessible phase-space volume, so counting optimized structures does not establish equilibrium populations.

2.4 Classical molecular dynamics

MD estimates equilibrium averages by integrating forces, with periodic boundaries and reservoir couplings appropriate to the ensemble.

2.4.1 Periodic boundaries and pressure

Periodic images differ by $\mathbf{h}\mathbf{n}$, $\mathbf{n} \in \mathbb{Z}^3$. A single-image cutoff requires r_c below half the shortest nonzero lattice-vector length throughout cell deformation; otherwise include all images within the cutoff. Skew cells require a geometric nearest-image search. Neighbor lists extend to $r_c + r_{\text{skin}}$ and must be rebuilt before omitted pairs can enter the cutoff. Smooth cutoffs avoid force jumps. Separate electrostatic terms, if present, need converged long-range treatment consistent with the potential.

For unconstrained atoms with bulk streaming motion removed,

$$\mathbf{P}_{\text{inst}} = \frac{1}{V} \left[\sum_i \frac{\mathbf{p}_i \otimes \mathbf{p}_i}{m_i} + \mathbf{W} \right], \quad P_{\text{inst}} = \frac{1}{3} \text{tr} \mathbf{P}_{\text{inst}}. \quad (46)$$

Here $\otimes$ denotes an outer product. Pair forces give $\mathbf{W} = \sum_{i < j} \mathbf{r}_{ij} \otimes \mathbf{F}_{ij}$; many-body potentials use strain derivatives at fixed fractional positions:

$$W_{\alpha\beta} = - \left. \frac{\partial U}{\partial \epsilon_{\alpha\beta}} \right|_{\epsilon=0, \{\mathbf{s}_i\}}, \quad \mathbf{h} \mapsto (\mathbf{I} + \boldsymbol{\epsilon})\mathbf{h}. \quad (47)$$

Include constraint and long-range contributions when present. Tensile-positive stress has the opposite sign to pressure; model virial omits the kinetic contribution to full pressure [7].

2.4.2 Time integration

Fixed-cell velocity Verlet updates momenta and positions as

$$\mathbf{p}_i^{n+1/2} = \mathbf{p}_i^n + \frac{\Delta t}{2} \mathbf{F}_i(\mathbf{R}^n), \quad (48)$$

$$\mathbf{R}_i^{n+1} = \mathbf{R}_i^n + \Delta t \frac{\mathbf{p}_i^{n+1/2}}{m_i}, \quad (49)$$

$$\mathbf{p}_i^{n+1} = \mathbf{p}_i^{n+1/2} + \frac{\Delta t}{2} \mathbf{F}_i(\mathbf{R}^{n+1}). \quad (50)$$

It is reversible, symplectic, and second order, requiring one new force evaluation per step. The harmonic stability condition $\omega_{\text{max}} \Delta t < 2$ does not ensure accurate sampling: converge the timestep using energy drift in short NVE tests and the target observables. Thermostats can conceal integration errors. This report retains flexible bonds; constraints would change the heat capacity and require matching harmonic mode bookkeeping.

2.4.3 Thermostats and barostats

Langevin dynamics samples the canonical distribution through balanced friction and Gaussian noise:

$$\dot{\mathbf{p}}_i = \mathbf{F}_i - \gamma \mathbf{p}_i + \boldsymbol{\eta}_i(t), \quad (51)$$

$$\langle \eta_{i\alpha}(t) \eta_{j\beta}(t') \rangle = 2m_i \gamma k_B T \delta_{ij} \delta_{\alpha\beta} \delta(t - t'). \quad (52)$$

Here γ is an inverse coupling time. Nosé–Hoover instead uses an auxiliary friction variable:

$$\dot{\mathbf{p}}_i = \mathbf{F}_i - \xi \mathbf{p}_i, \quad \dot{\xi} = \frac{2K - f k_B T}{Q}, \quad (53)$$

The inertia $Q \sim f k_B T \tau_T^2$ sets the response time. Chains improve sampling of nearly harmonic systems [22], while simple weak coupling or exact velocity rescaling can suppress required fluctuations.

A barostat changes volume or cell strain. Flexible-cell MTK algorithms include the measure corrections required for NPT [21]. Overly rapid coupling excites cell motion; slow coupling limits exploration. Converge coupling times and check pressure averages and fluctuations. Thermostatted trajectories yield equilibrium properties, but transport coefficients also require a validated dynamical protocol.

Table 1: MD stages and their acceptance checks. Actual production settings appear in Section 4.3.

Stage	Required check
Define system	Composition, ensemble, periodic cell, masses, units, model domain, and validated forces/stresses.
Prepare replicas	Independent guest placements and velocities; reject overlaps and inspect initial forces.
Validate integration	NVE energy conservation; timestep, neighbor-list, and coupling-time convergence.
Equilibrate	Stationary enthalpy, cell, framework, and guest distributions; compatible replicas.
Collect and analyze	Save physical observables and restarts; discard transients, estimate correlations, and propagate derivative errors.
Converge and archive	System size, duration, replica count, temperature spacing, model identity, seeds, and analysis provenance.

2.4.4 Simulation workflow

2.5 Graph theory for neural networks and sparse differentiation

Graphs describe atomic neighborhoods and derivative dependencies [13, 15]. Their roles are distinguished in Table 2.

2.5.1 Adjacency, paths, and receptive fields

A graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ contains vertices and edges. Atomic graphs are directed periodic multigraphs: messages have directions, and repeated images can connect the same atom labels. Edges indicate interactions, not necessarily chemical bonds. For a simple undirected graph,

$$A_{ij} = \begin{cases} 1, & \{i, j\} \in \mathcal{E}, \\ 0, & \text{otherwise,} \end{cases} \quad \mathcal{N}_i = \{j : A_{ij} = 1\}. \quad (54)$$

The degree is $d_i = |\mathcal{N}_i|$. Species and displacements add node and edge attributes; Boolean adjacency alone loses geometry and image multiplicity.

Graph distance counts edges in a shortest path. Its k -hop ball is

$$\mathcal{B}_k(i) = \{j : d_{\mathcal{G}}(i, j) \leq k\}. \quad (55)$$

For $1-2-3-4$, $\mathcal{B}_1(1) = \{1, 2\}$ and $\mathcal{B}_2(1) = \{1, 2, 3\}$. Boolean powers of adjacency record reachability (Eq. (106)). With L local message-passing layers, a node depends on its L -hop neighborhood. Atoms influencing the same nonlinear readout can have mixed second derivatives beyond one hop.

Relabeling by $\mathbf{P}$ transforms adjacency to $\mathbf{PAP}^T$. Node outputs must permute consistently; total energy must remain invariant (Eq. (66)). The computation graph instead records operations and directed dependencies. Forward evaluation/AD follows its acyclic order; reverse AD applies the chain rule backward [26].

Table 2: Graph constructions used by the potential and its differentiation.

Graph	Vertices	Meaning of an edge
Interaction	Atoms	Neighbor relation, with periodic-image attributes
Computation	Values or operations	Directed computational dependence
Hessian sparsity	Coordinates (or atom blocks)	Possible off-diagonal curvature coupling
Column intersection	Derivative columns	Possible nonzeros sharing an output row

2.5.2 Coloring derivative directions

A sparsity pattern $S_{pq} = 1$ permits dependence of output p on input q . Columns sharing a possible nonzero row are adjacent in the column-intersection graph. Proper coloring assigns them different colors, so same-color columns have disjoint supports and can share an AD seed [27].

$$\mathbf{M} = \begin{pmatrix} a & b & 0 & 0 \\ b & c & d & 0 \\ 0 & d & e & f \\ 0 & 0 & f & g \end{pmatrix}. \quad (56)$$

For this tridiagonal matrix, alternating chain colors $(1, 2, 1, 2)$ fail: columns 1 and 3 both contribute to row 2. Colors $(1, 2, 3, 1)$ allow columns 1 and 4 to share a seed:

$$\mathbf{M}(1, 0, 0, 1)^\top = (a, b, f, g)^\top. \quad (57)$$

Three products then recover the matrix instead of four.

A *star coloring* of the symmetric sparsity graph requires at least three colors on every four-vertex path; every two-color subgraph is a union of stars. The chain coloring $(1, 2, 1, 3)$ satisfies this condition. Hessian symmetry provides a second compressed location for each off-diagonal entry, allowing some overlapping supports. SADMOF uses this recovery strategy [14, 28]; Section 4.4.2 gives the formulas. Either strategy is exact only when the pattern includes every actual coupling.

2.6 Machine-learned interatomic potentials

2.6.1 Energy, forces, and atomic graphs

An MLIP approximates the Born–Oppenheimer map $(\mathbf{R}, \mathbf{h}, \mathbf{Z}) \mapsto U$ with fitted parameters θ . A local, extensive decomposition is

$$U_\theta(\mathbf{R}, \mathbf{h}, \mathbf{Z}) = U_{\text{base}}(\mathbf{Z}) + \sum_{i=1}^N \varepsilon_i(\mathcal{N}_i; \theta), \quad (58)$$

The optional baseline depends only on composition. Atomic energies are model contributions, not unique observables. At bounded neighbor count, evaluation cost scales approximately linearly with atom count. Differentiating the same scalar energy gives conservative forces and a Cartesian Hessian:

$$\mathbf{F}_i = -\frac{\partial U_\theta}{\partial \mathbf{R}_i}, \quad (59)$$

Atoms are graph vertices with species features. Periodic directed edges $(v, u, \mathbf{n})$ carry relative geometry:

$$\mathbf{r}_{uv\mathbf{n}} = \mathbf{R}_v + \mathbf{h}\mathbf{n} - \mathbf{R}_u, \quad (60)$$

$$r_{uv\mathbf{n}} = \|\mathbf{r}_{uv\mathbf{n}}\| < r_c, \quad (61)$$

$$\mathcal{N}_u = \{(v, \mathbf{n}) : (v, u, \mathbf{n}) \in \mathcal{E}\}. \quad (62)$$

Retaining image labels distinguishes repeated neighbors. Species embeddings and radial edge features provide numerical inputs:

$$\mathbf{x}_u^{(0)} = \mathbf{E}_Z[Z_u] \in \mathbb{R}^d, \quad (63)$$

$$\mathbf{x}_{uv\mathbf{n}}^{\text{edge}} = f_c(r_{uv\mathbf{n}})[\varphi_1(r_{uv\mathbf{n}}), \dots, \varphi_K(r_{uv\mathbf{n}})], \quad (64)$$

Directional models also use displacement vectors or their learned functions; latent feature vectors need not transform as physical vectors.

Figures 2 and 3 illustrate cutoff weighting. Smooth forces and Hessians require consistent derivatives through attention normalization and readout as well as radial features.

Node-wise F and scalar readout f obey

$$F(\mathbf{P}\mathbf{X}, \mathbf{P}\mathbf{A}\mathbf{P}^\top) = \mathbf{P}F(\mathbf{X}, \mathbf{A}), \quad (65)$$

$$f(\mathbf{P}\mathbf{X}, \mathbf{P}\mathbf{A}\mathbf{P}^\top) = f(\mathbf{X}, \mathbf{A}). \quad (66)$$

Relabel edge attributes and periodic images consistently.

2.6.2 Message passing

Each layer combines incoming messages with a symmetric aggregation $\bigoplus$ (usually a sum), then updates the node representation [15]:

$$\mathbf{h}_u = \phi \left(\mathbf{x}_u, \bigoplus_{(v, \mathbf{n}) \in \mathcal{N}_u} \psi(\mathbf{x}_u, \mathbf{x}_v, \mathbf{x}_{uv\mathbf{n}}^{\text{edge}}) \right). \quad (67)$$

Repeating the update through L layers creates many-body features over an L -hop receptive field. A shared readout $\varepsilon_u = \rho(\mathbf{h}_u^{(L)})$ gives the total energy through Eq. (58).

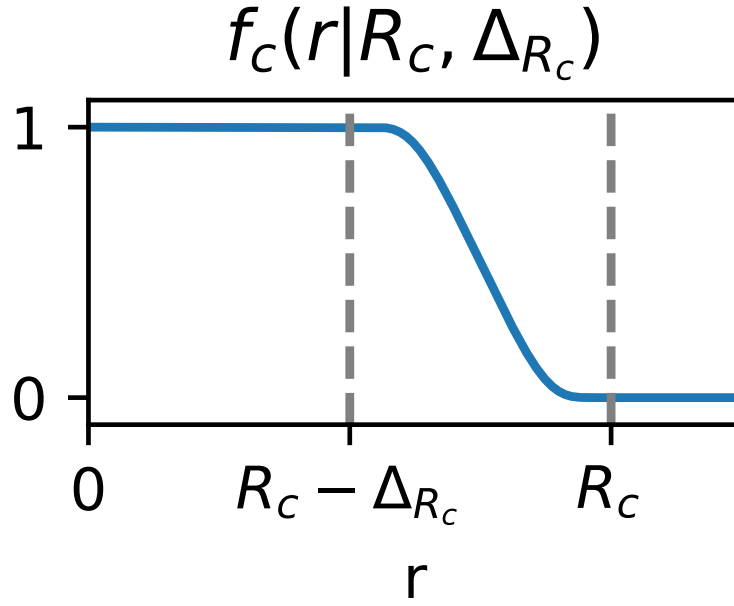


Figure 2: A smooth radial cutoff suppresses a neighbor contribution as its distance approaches the outer cutoff r_c . Smooth decay is needed for continuous energies, forces, and Hessians. Figure from the PET paper [19].

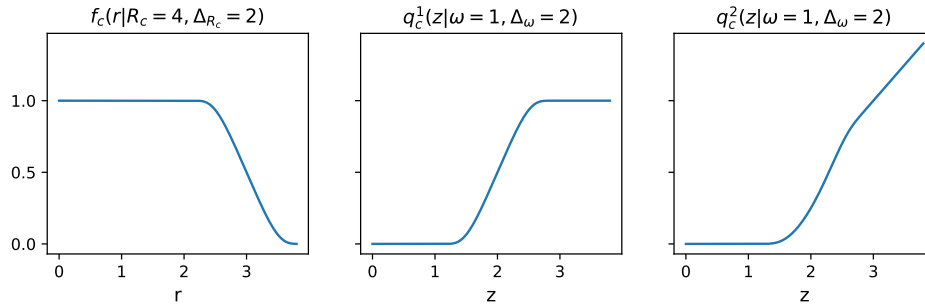


Figure 3: Comparison of radial cutoff functions used to attenuate neighbor contributions near r_c . The value and the derivatives relevant to the model must be treated consistently in every energy contribution. Figure from the PET paper [19].

2.6.3 Point Edge Transformer architecture

PET retains distinct directed-edge messages while attention mixes all tokens in one atomic environment (Figure 4) [19]. It underlies PET-MAD [18]. In the original formulation,

$$\mathbf{p}_{ij}^{(\ell)} = \text{SiLU}(\mathbf{W}_r^{(\ell)} \mathbf{r}_{ij} + \mathbf{b}_r^{(\ell)}), \quad (68)$$

$$\mathbf{t}_{ij}^{(\ell)} = \text{MLP}_\ell(\mathbf{m}_{j \rightarrow i}^{(\ell-1)} \parallel \mathbf{E}_n^{(\ell)}[Z_j] \parallel \mathbf{p}_{ij}^{(\ell)}), \quad (69)$$

Here $\parallel$ concatenates the incoming message, neighbor embedding, and position encoding. The MLP maps $3d_{\text{PET}}$ features to d_{PET} ; the first block omits the incoming message. A separate central token is $\mathbf{t}_{i0}^{(\ell)} = \mathbf{E}_c^{(\ell)}[Z_i]$.

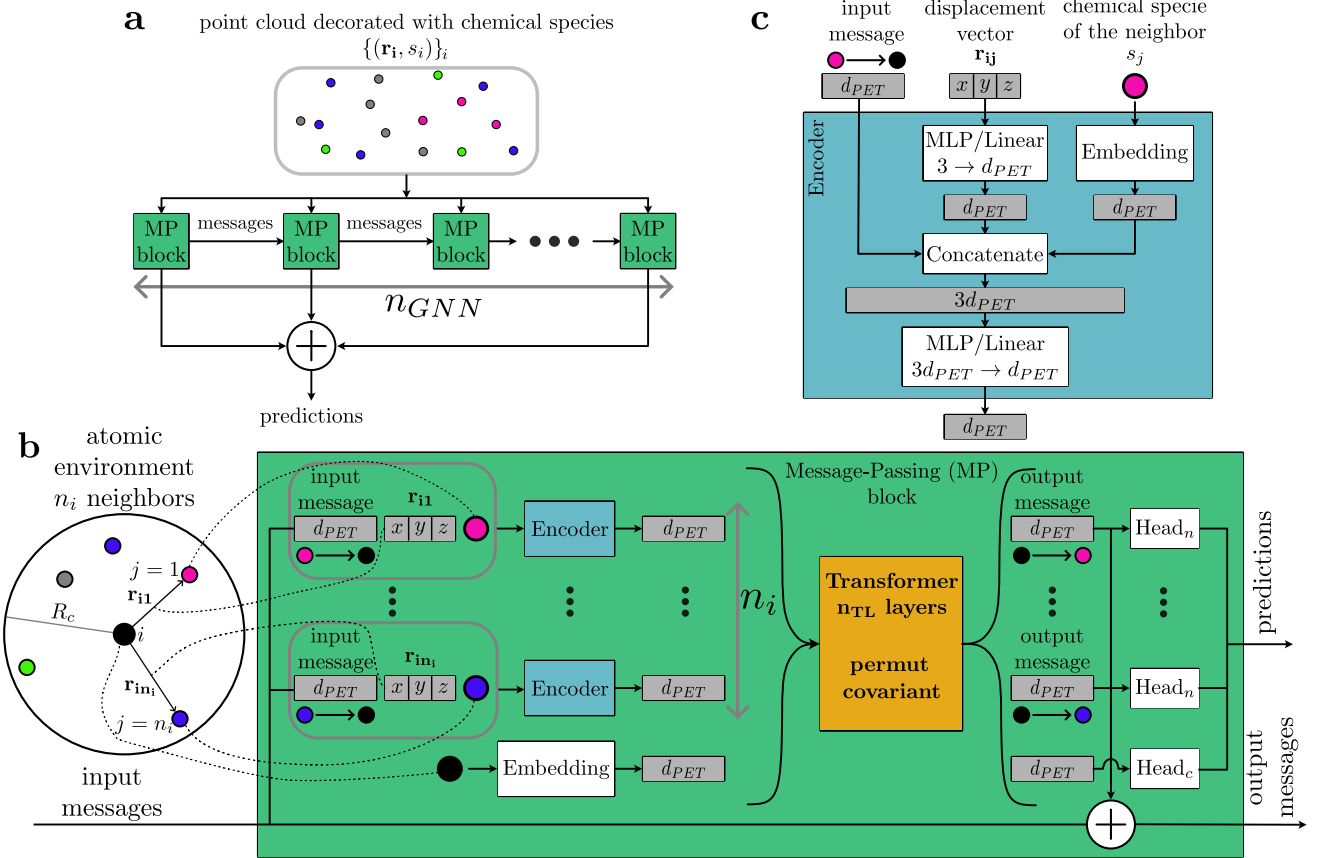


Figure 4: Point Edge Transformer (PET) message-passing architecture. Edge messages associated with a central atom are jointly processed by attention while retaining their individual edge identities for the outgoing update. Figure from the PET paper [19].

For one attention head,

$$\mathbf{q}_a = \mathbf{W}_Q \mathbf{t}_a, \quad \mathbf{k}_b = \mathbf{W}_K \mathbf{t}_b, \quad (70)$$

$$\mathbf{v}_b = \mathbf{W}_V \mathbf{t}_b, \quad (71)$$

$$\alpha_{ab} = \frac{\exp(\mathbf{q}_a^T \mathbf{k}_b / \sqrt{d_k}) c_b}{\sum_c \exp(\mathbf{q}_a^T \mathbf{k}_c / \sqrt{d_k}) c_c}, \quad (72)$$

$$\mathbf{y}_a = \sum_b \alpha_{ab} \mathbf{v}_b. \quad (73)$$

With $c_0 = 1$ and $c_j = f_c(r_{ij})$, departing neighbors vanish from both numerator and normalization. Multi-head attention, feed-forward layers, and residual connections preserve neighbor permutation equivariance. Attention layers enrich interactions within one environment; message-passing blocks extend the receptive field. Local attention cost is quadratic in neighbor count, but total cost remains approximately linear in atom count at fixed architecture and bounded neighborhoods.

Outgoing edge tokens update the messages. The original summed-edge readout is

$$U_\theta = \sum_i b_{Z_i} + \sum_{\ell=1}^L \sum_i \left[H_c^{(\ell)}(\mathbf{x}_{i0}^{(\ell)}) + \sum_{j \in \mathcal{N}_i} f_c(r_{ij}) H_n^{(\ell)}(\mathbf{x}_{ij}^{(\ell)}) \right], \quad (74)$$

H_c and H_n are learned heads, $\mathbf{x}$ are output tokens, and b_Z are species baselines. Every block contributes; cutoff weighting also enters the readout. Edge contributions are learned many-body terms, not pair energies. Project-specific dimensions appear in Section 4.2.

2.6.4 Physical symmetries and ECSE

Scalar energies should respect translation, rotation/reflection, and identical-atom permutation:

$$\begin{aligned} U(\mathbf{R}) &= U(\mathbf{R} + \mathbf{a}) = U(\mathbf{Q}\mathbf{R}) \\ &= U(\mathbf{P}\mathbf{R}), \quad \mathbf{Q} \in O(3). \end{aligned} \quad (75)$$

Forces then transform equivariantly, $\mathbf{F}_i(\mathbf{Q}\mathbf{R}) = \mathbf{Q}\mathbf{F}_i(\mathbf{R})$; periodic tests rotate both positions and cell. PET enforces translation and permutation symmetries, but learns rotational behavior through augmentation. Reported rotational errors below permutation errors [18] do not validate this checkpoint on loaded MOF-5.

The optional Equivariant Coordinate System Ensemble (ECSE) imposes exact rotation symmetry through local frames from ordered noncollinear neighbor pairs [19]:

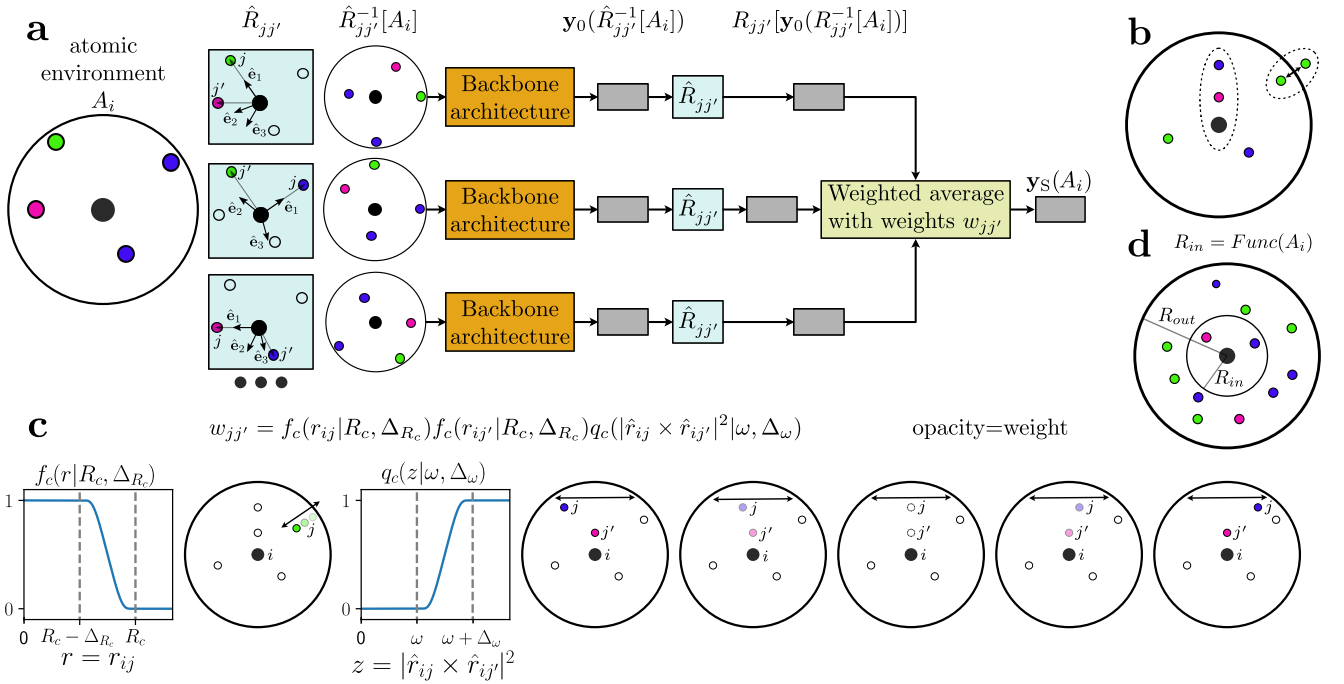


Figure 5: Equivariant Coordinate System Ensemble (ECSE). Local coordinate frames constructed from non-collinear neighbor pairs rotate with the atomic environment; predictions evaluated in these frames are transformed as needed and smoothly averaged. Figure from the PET paper [19].

$$\varepsilon_i^{\text{ECSE}} = \frac{\sum_{j \neq k} w_{ijk} \varepsilon_i^0(\mathbf{Q}_{ijk}^T \mathcal{N}_i)}{\sum_{j \neq k} w_{ijk}}, \quad (76)$$

$\mathbf{Q}_{ijk}^T \mathcal{N}_i$ expresses displacements in a co-rotating frame; vector/tensor outputs are transformed back before averaging. These proper $SO(3)$ rotations alone do not enforce reflection invariance. Smooth radial/angular weights suppress cutoff and nearly collinear frames; fully collinear environments need a fallback. The full frame and weight dependence must be differentiated for forces and Hessians. Tight frame selection can amplify derivative errors, and the proof-of-principle overhead is roughly 10^3 [19]. ECSE is not enabled here.

2.6.5 Training and validity

A generic energy–force objective is

$$\mathcal{L}(\theta) = \sum_s \left[w_E \left(\frac{U_\theta^{(s)} - U_{\text{ref}}^{(s)}}{N_s} \right)^2 + \frac{w_F}{3N_s} \sum_{i\alpha} \left(F_{i\alpha,\theta}^{(s)} - F_{i\alpha,\text{ref}}^{(s)} \right)^2 \right], \quad (77)$$

Stress terms and regularization may be added. PET’s original loss uses a different normalization with moving validation errors, species baselines, Adam, and rotational augmentation [19]. PET-MAD is pretrained on a chemically diverse DFT dataset [18].

An MLIP inherits its reference method’s approximations and may extrapolate poorly to unfamiliar environments. Hold-out errors, trajectory stability, and uncertainty indicators test different aspects of validity. Curvatures need explicit validation because the usual loss does not constrain Hessians directly; PET’s methane benchmarks do not establish accuracy for adsorption in MOF-5.

2.7 Metal–organic frameworks and MOF-5

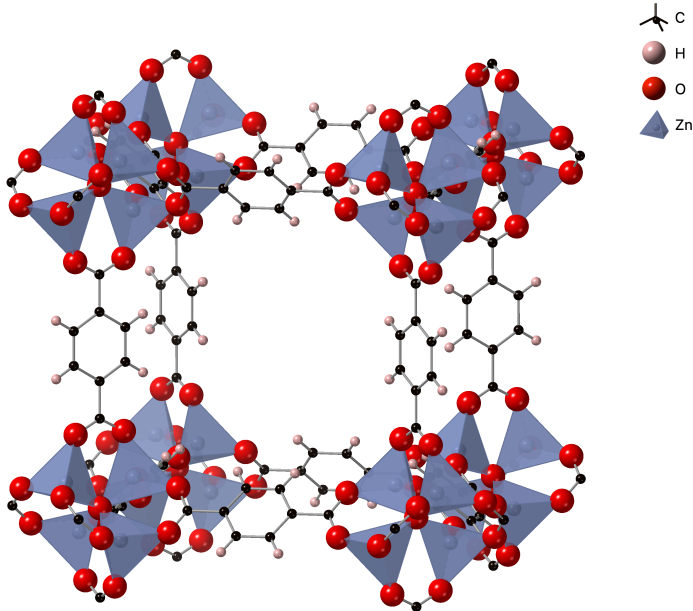


Figure 6: Ball-and-polyhedron representation of the MOF-5 framework. The zinc–oxo clusters are shown as blue polyhedra, connected by benzene-1,4-dicarboxylate linkers. The legend identifies carbon (black), hydrogen (pink), oxygen (red), and zinc (blue). The open spaces between framework components form the pores that host adsorbed molecules.

MOF-5 consists of zinc–oxo nodes joined by benzene-1,4-dicarboxylate linkers (Figure 6). Methane occupies the pores and interacts mainly through weak noncovalent forces. Framework vibration, guest internal modes, and redistribution between adsorption environments all contribute to heat capacity. Loaded-system MD captures the anharmonic motion; its optimized normal modes supply the quantum correction [16].

2.8 Thermodynamic definitions and uncertainty

The classical term differentiates physical NPT enthalpy at fixed loading and pressure. The harmonic correction is formally C_V ; adding it to C_P assumes a small quantum pressure–volume contribution (Eq. (37)). Combine extensive terms before dividing by loaded mass.

2.8.1 LLPR model uncertainty and CEA propagation

Finite-trajectory sampling error is distinct from potential uncertainty. LLPR varies the final readout of a fixed PET representation [23]. For central energy $\bar{U} = \mathbf{w}_0^T \mathbf{f}(\mathbf{x})$ and covariance-set features $\mathbf{F}$,

$$\sigma_{\text{LLPR}}^2(\mathbf{x}) = \alpha^2 \mathbf{f}(\mathbf{x})^T \times (\mathbf{F}^T \mathbf{F} + \lambda \mathbf{I})^{-1} \mathbf{f}(\mathbf{x}). \quad (78)$$

The regularizer λ stabilizes weakly constrained directions; α is calibrated against energy residuals on a disjoint reference-labeled set using the base potential's electronic-structure definition. Predictions cannot calibrate their own error.

The exported `energy_uncertainty` is an energy standard deviation. Property propagation instead needs persistent, signed energy variations:

$$\begin{aligned} \mathbf{w}_i &\sim \mathcal{N}[\mathbf{w}_0, \alpha^2 (\mathbf{F}^T \mathbf{F} + \lambda \mathbf{I})^{-1}], \\ U^{(i)}(\mathbf{x}) &= \mathbf{w}_i^T \mathbf{f}(\mathbf{x}), \end{aligned} \quad (79)$$

The exported `energy_ensemble` preserves member identity across configurations. Its large-ensemble spread approaches the analytical LLPR value, but cannot capture bias shared by the frozen representation.

Following Imbalzano *et al.* [25], central-trajectory frames supply

$$\Delta U_t^{(i)} = U_t^{(i)} - \bar{U}_t, \quad (80)$$

$$H_t^{(i)} = K_t + U_t^{(i)} + P_{\text{ext}} V_t, \quad (81)$$

$$\langle H^{(i)} \rangle_i^{\text{direct}} = \frac{\sum_t e^{-\beta \Delta U_t^{(i)}} H_t^{(i)}}{\sum_t e^{-\beta \Delta U_t^{(i)}}}, \quad (82)$$

Extensive energy differences can collapse the exponential weights. The primary large-system estimate uses first-order cumulant expansion (CEA):

$$\begin{aligned} \langle H^{(i)} \rangle_i^{\text{CEA}} &= \langle H^{(i)} \rangle_{\bar{U}} \\ &\quad - \beta \text{cov}_{\bar{U}}(H^{(i)}, \Delta U^{(i)}), \\ \beta &= (k_B T)^{-1}. \end{aligned} \quad (83)$$

Direct reweighting remains a diagnostic: $N_{\text{eff}} = 1/\sum_t \tilde{w}_t^2$ should remain adequate, and $\text{var}_t(\beta \Delta U_t^{(i)}) \ll 1$ is needed for controlled first-order CEA.

Average replicas member by member, preserve indices across temperatures, and differentiate each complete enthalpy curve:

$$\begin{aligned} C_{P,\text{cl}}^{(i)}(T) &= \frac{d\langle H^{(i)} \rangle_i^{\text{CEA}}}{dT}, \\ \sigma_{\text{LLPR}, C_P}(T) &= \left\{ \frac{1}{M-1} \sum_{i=1}^M \left[C_{P,\text{cl}}^{(i)}(T) - \bar{C}_{P,\text{cl}}(T) \right]^2 \right\}^{1/2}. \end{aligned} \quad (84)$$

The final property spread is a predictive standard deviation; do not divide it by $\sqrt{M}$.

2.8.2 Gaussian NPT propagation

Direct propagation of shallow ensembles (DPOSE) combines member properties before measuring their spread [24]. The saved comparison uses a Gaussian NPT approximation on central-model trajectories. Define $Q_i = U_i + P_{\text{ext}} V$ and $\Delta U_i = U_i - U_0$. Joint Gaussianity leaves covariance unchanged under exponential reweighting, giving

$$C_{P,i}^G \simeq \frac{f}{2} k_B + \frac{\text{var}_0(Q_i)}{k_B T^2}. \quad (85)$$

Here $f = 3N - 3$. The central-model fluctuation estimator is exact under the usual equilibrium assumptions; using central-trajectory variances for other members needs Gaussian closure. Pair member indices with harmonic corrections.

The main Gaussian plots replace only the propagated model spread around the central enthalpy-derivative curve. The separate variance-based hybrid also replaces the central classical value and estimates sampling error from squared configurational-enthalpy fluctuations. Neither guarantees smaller or better calibrated uncertainty.

2.8.3 Classical versus quantum vibrations

Classical harmonic modes contribute k_B each at every temperature. Quantum occupation decreases when $\hbar\omega \geq k_B T$ (Eq. (119)). Longer classical trajectories cannot recover that suppression.

2.9 Path integral molecular dynamics

PIMD samples the quantum nuclear statistics of Section 2.1 on the full anharmonic Born–Oppenheimer surface. We will use it to assess quantum effects in water-loaded MOFs, particularly those associated with hydrogen. Present methane results use classical MD plus a harmonic correction. The following assumes distinguishable nuclei and neglects exchange permutations [2].

2.9.1 Ring-polymer mapping

Starting from Eq. (8) at fixed cell, write $\hat{\mathcal{H}} = \hat{\mathcal{H}}_{\text{BO}} = \hat{K} + \hat{U}$. The noncommuting nuclear kinetic and potential operators are split into L imaginary-time slices, $\beta_L = \beta/L$:

$$e^{-\beta\hat{\mathcal{H}}} = \lim_{L \rightarrow \infty} \left[e^{-\beta_L \hat{U}/2} e^{-\beta_L \hat{K}} e^{-\beta_L \hat{U}/2} \right]^L. \quad (86)$$

Inserting position states gives

$$Z_L = \mathcal{A}_L \int \prod_{j=1}^L d^{3N} \mathbf{R}^{(j)} e^{-\beta U_L^{\text{eff}}}, \quad (87)$$

$$\mathcal{A}_L = \prod_{i=1}^N \left(\frac{m_i L}{2\pi\beta\hbar^2} \right)^{3L/2}, \quad (88)$$

$$U_L^{\text{eff}} = \frac{1}{L} \sum_{j=1}^L U(\mathbf{R}^{(j)}; \mathbf{h}) + \frac{1}{L} \sum_{i,j} \frac{m_i \omega_L^2}{2} |\mathbf{R}_i^{(j)} - \mathbf{R}_i^{(j+1)}|^2, \quad (89)$$

$$\omega_L = \frac{L}{\beta\hbar}, \quad \mathbf{R}_i^{(L+1)} = \mathbf{R}_i^{(1)}. \quad (90)$$

Each nucleus becomes a closed ring of L beads connected by springs; the physical potential acts within each slice. Beads resolve imaginary time, not a real trajectory. Periodic paths need consistent images and, where relevant, winding sectors. The equivalent convention samples LU_L^{eff} at LT . At $L = 1$ the classical configuration weight is recovered; the symmetric primitive error is $\mathcal{O}(L^{-2})$ for smooth potentials [2, 9].

2.9.2 Sampling and observables

Auxiliary momenta sample

$$\mathcal{H}_{\text{samp}} = \sum_{i,j} \frac{|\boldsymbol{\pi}_i^{(j)}|^2}{2\mu_i^{(j)}} + U_L^{\text{eff}}. \quad (91)$$

Fictitious masses μ affect efficiency; physical masses m in the springs determine quantum delocalization. Normal-mode thermostats such as PILE handle stiff springs [9]. PIMD simulation time is only a sampling variable; quantum transport or rates require additional real-time approximations such as RPMD/CMD.

For position-diagonal observables,

$$\langle A \rangle_{\text{q}} = \lim_{L \rightarrow \infty} \left\langle \frac{1}{L} \sum_{j=1}^L A(\mathbf{R}^{(j)}) \right\rangle_L, \quad (92)$$

A nonlinear observable at the centroid generally differs from the bead average. A centroid-virial physical energy estimator is

$$\bar{\mathbf{R}}_i = \frac{1}{L} \sum_j \mathbf{R}_i^{(j)}, \quad (93)$$

$$K_{\text{cv}} = \frac{3N}{2\beta} - \frac{1}{2L} \sum_{i,j} (\mathbf{R}_i^{(j)} - \bar{\mathbf{R}}_i) \cdot \mathbf{F}_i^{(j)}, \quad (94)$$

$$E_{\text{est}} = K_{\text{cv}} + \frac{1}{L} \sum_j U(\mathbf{R}^{(j)}; \mathbf{h}), \quad (95)$$

Use physical forces and unwrapped paths, with matching modifications for constraints or removed degrees of freedom [10]. NPT also requires a shared cell, quantum pressure estimator, and correct cell measure [11]:

$$\langle H \rangle_q = \lim_{L \rightarrow \infty} \langle E_{\text{est}} + P_{\text{ext}} V \rangle_L, \quad (96)$$

$$C_P^q = \left(\frac{\partial \langle H \rangle_q}{\partial T} \right)_{N, P_{\text{ext}}}. \quad (97)$$

Exclude fictitious kinetic and spring energies from physical enthalpy. A mean-energy estimator’s variance is not automatically the quantum enthalpy variance; direct heat-capacity estimators must differentiate the action, normalization, and estimator [10].

2.9.3 Prospective validation

Converge beads at every derivative temperature, starting with $\beta \hbar \omega_{\text{max}}/L \ll 1$. Each step costs approximately L force evaluations. An i-PI/PET interface [12] should first recover $L = 1$ classical results, then stable quantum harmonic benchmarks, before loaded-MOF NPT tests. Converge timestep, thermostats, cell size, and sampling as well as beads; validate quantum-displaced configurations against the potential’s domain. For H_2O loadings, test O–H vibrations and hydrogen-bond observables alongside heat capacity; high-frequency hydrogen modes can require more beads even near room temperature. Converged PIMD would test the hybrid’s missing quantum anharmonic and pressure–volume terms. Do not add a harmonic quantum correction again to PIMD results.

3 Reference evidence and the hybrid approximation

Kapil *et al.* tested empty MOF-5 and 50-, 100-, and 150-methane cells [16]. Their hybrid strategy motivates this project; its accuracy on PET remains to be established.

3.1 Differences from the reference calculation

The reference used QuickFF [20]: B3LYP cluster-based bonded terms, MBIS charges from periodic PBE, Gaussian electrostatics, and MM3 Buckingham interactions with a 15 Å cutoff and tail corrections. RASPA insertion and canonical Monte Carlo preceded MD. Here PET supplies all energies, forces, and Hessians after independent random guest placement.

Both workflows differentiate neighboring equilibrium enthalpies:

$$C_P(T) \simeq \frac{\langle H \rangle_{T+\Delta T} - \langle H \rangle_{T-\Delta T}}{2\Delta T}, \quad (98)$$

$$\Delta T = 25 \text{ K},$$

The reference used optimized Yaff Hessians and TAMkin modes, compared QuickFF and UFF4MOF for the empty host, and optimized five Monte Carlo configurations for the 100-methane harmonic calculation.

3.2 Physical conclusions

The reference found nearly harmonic, quantum-mechanical empty MOF-5 (roughly 0.7 J/g/K near room temperature). Classical occupations overestimate its heat capacity. Methane introduces anharmonic host–guest motion and a loading-dependent minimum near 200 K, associated with changing attractive interactions as guests become mobile. The combination

$$C \approx C_{\text{qn}}^{\text{har}} - C_{\text{cl}}^{\text{har}} + C_{\text{cl}}^{\text{anh}} \quad (99)$$

reproduced the minimum qualitatively and was useful above about 200 K in their 100-methane test. A loaded-system spectrum is essential: the empty Hessian omits guest modes and host–guest coupling. Equation (123) gives the operational PET estimator.

4 Present computational methodology

4.1 Structure preparation and methane insertion

The conventional host has 424 atoms and cell parameter 25.866 Å. The campaign module `mof_heat_capacity_protocols.loaded` inserts centered methane molecules with uniform quaternion rotations and fractional positions, rejecting periodic atom–atom overlaps. Independent insertion and velocity seeds define replicas. Random placement prevents severe contacts but is not an equilibrium adsorption distribution; subsequent equilibration must be validated.

4.2 PET model and implementation

The supplied PET-MAD and PET-SOL conversions share the architecture in Table 3; identical settings do not imply identical weights or training data. The original PET default used three attention layers per block, rather than one here.

Table 3: Architecture settings recorded in the local PET-MAD and PET-SOL PET-JAX metadata. The species count is embedding capacity, not a measure of training coverage.

Setting	Value
Outer cutoff	8.0 Å
Cutoff switching width	0.5 Å
Adaptive neighbor-selection setting	40
Message-passing blocks	3
Attention layers per block	1
Token width, d_{PET}	256
Attention heads	8
Transformer feed-forward width	512
Node feature width, d_node	1024
Readout feature width, d_head	256
Species-embedding capacity	102

The installed PyTorch implementation (`metatrain 2026.3.1`) embeds $(x_{ij}, y_{ij}, z_{ij}, r_{ij})$ and initializes edge messages with species embeddings. Central features contract to token width for attention and expand afterward. Attention-head width is $256/8 = 32$; **d_head=256** denotes readout width. Neighbor masking and log-cutoff logits implement Eq. (73). Reverse-neighbor routing and averaging with old messages update edge states.

The repository exports checkpoints with `upet.save_upet` for LAMMPS/metatomic MD, and converts matching weights to PET-JAX for SADMOF Hessians. Setup pins PET-JAX to commit `ed0add4`; conversion does not retrain. Composition baselines have zero Cartesian derivatives; energy scales must carry through forces and Hessians.

Compare energies and forces across export and conversion, including adaptive-cutoff derivative conventions. The sparse path stops those derivatives (Section 4.4.2). NPT additionally requires validated stress output on representative loaded-MOF structures; matching filenames or capability flags alone are insufficient [18].

4.3 Classical NPT molecular dynamics

Only loaded MOF-5 is propagated. LAMMPS/metatomic uses a flexible triclinic cell with isotropic 1 bar target pressure and a Nosé–Hoover-chain/MTTK NPT fix. The campaign settings are

Table 4: Configured NPT campaign. Equilibration and convergence are assessed from observables, not assumed from these durations.

Parameter	Value
Thermostat chain length	3
Thermostat / barostat damping	100 / 1000 fs
Timestep / total steps	0.5 fs / 10^6
Trajectory / discarded interval	500 / 100 ps
Save interval	100 steps (0.05 ps)
Models / methane loadings	PET-MAD, PET-SOL / 50, 100, 150
MD grid	175–425 K, every 25 K
Hybrid reporting grid	200–400 K, every 25 K
Available replicas	One per state point

Production means use physical enthalpy,

$$\langle H \rangle_T = \langle E + P_{\text{ext}} V \rangle_T, \quad (100)$$

then replica averaging, where available, and centered differentiation with neighbors separated by 50 K. This term retains sampled anharmonic motion. Independent replicas, longer production, and temperature-spacing convergence remain outstanding (Section 5.2).

4.4 Harmonic approximation and Cartesian Hessian

Let $\mathbf{R}_0$ be a stationary reference structure and $\mathbf{u} = \mathbf{R} - \mathbf{R}_0$. Expanding the potential energy gives

$$U(\mathbf{R}_0 + \mathbf{u}) = U(\mathbf{R}_0) + \underbrace{\nabla U(\mathbf{R}_0) \cdot \mathbf{u}}_{=0 \text{ at a minimum}} + \frac{1}{2} \mathbf{u}^T \mathbf{H} \mathbf{u} + \mathcal{O}(u^3), \quad (101)$$

with Cartesian Hessian

$$H_{i\alpha, j\beta} = \frac{\partial^2 U}{\partial R_{i\alpha} \partial R_{j\beta}} = -\frac{\partial F_{i\alpha}}{\partial R_{j\beta}}. \quad (102)$$

For N atoms, $\mathbf{H}$ has dimensions $3N \times 3N$; for the 100-methane cell this is 2772×2772 .

4.4.1 Automatic differentiation

At fixed parameters, a neural-network energy composes differentiable operations:

$$\mathbf{z}^{(0)} = \mathbf{R}, \quad \mathbf{z}^{(\ell+1)} = \phi_\ell(\mathbf{z}^{(\ell)}; \theta), \quad U_\theta = \rho(\mathbf{z}^{(L)}). \quad (103)$$

AD applies their chain rules without a finite-displacement step [26]. It differentiates the implemented potential, up to floating-point error; it does not improve the potential's accuracy. For $\mathbf{f} : \mathbb{R}^n \rightarrow \mathbb{R}^m$,

$$(\mathbf{J}_f)_{pq} = \frac{\partial f_p}{\partial x_q}, \quad \mathbf{J}_f \in \mathbb{R}^{m \times n}. \quad (104)$$

Forward AD evaluates $\mathbf{J}_f \mathbf{v}$ (JVP); reverse AD evaluates $\mathbf{w}^T \mathbf{J}_f$ (VJP). Dense recovery requires n forward or m reverse seeds, without storing layer Jacobians [27]. A single reverse sweep yields ∇U and hence all forces. Forward-over-reverse evaluates

$$\mathbf{H} \mathbf{v} = \left. \frac{d}{d\epsilon} \nabla_{\mathbf{R}} U_\theta(\mathbf{R} + \epsilon \mathbf{v}) \right|_{\epsilon=0}. \quad (105)$$

Dense Hessian recovery needs $3N$ such products. Sparse reconstruction reduces the seed count; cutoff smoothness and neighbor-selection conventions still control which function is differentiated.

4.4.2 Sparse Hessian reconstruction with SADMOF

SADMOF constructs an atom-pair pattern from PET's selected-neighbor graph, then expands each pair into a Cartesian 3×3 block [28]. For L message-passing layers, the maximum Hessian reach is $K = 2L$ for node readout and $K = 2L + 1$ for PET edge readout. The paper's MACE-MP-0 medium, PET-XS, and PET-S reaches are four, five, and seven hops. Boolean adjacency gives

$$\mathbf{P}^{(k)} = \mathbf{I} \vee \mathbf{A} \vee \mathbf{A}^2 \vee \dots \vee \mathbf{A}^k, \quad (106)$$

At $k = K$ the pattern contains all couplings under the chosen derivative convention; $k < K$ truncates it. Additional nonlocal terms need separate analysis. Structural sparsity requires

$$S_{pq} = 0 \implies H_{pq} = 0 \quad (107)$$

An overestimate costs entries and colors. An underestimate can both discard curvature and contaminate retained entries. Mixed second derivatives depend on nonlinear readout, not just direct interaction edges.

For direct column recovery, connect columns with overlapping supports:

$$(q, r) \in \mathcal{E}_S \iff \exists p : S_{pq} = S_{pr} = 1. \quad (108)$$

A proper coloring makes same-color columns structurally orthogonal. Greedy heuristics need not minimize color count to recover exact entries [27]. Define one seed per color class,

$$\mathbf{v}_a = \sum_{q \in \mathcal{C}_a} \mathbf{e}_q, \quad \mathbf{H} \mathbf{v}_a = \sum_{q \in \mathcal{C}_a} \mathbf{H} \mathbf{e}_q. \quad (109)$$

and collect seeds in $\mathbf{V} \in \{0, 1\}^{3N \times n_c}$:

$$\mathbf{Y} = \mathbf{H} \mathbf{V}, \quad Y_{pa} = \sum_{q: c(q)=a} H_{pq}. \quad (110)$$

Disjoint supports permit

$$H_{pq} = Y_{p,c(q)} \quad \text{when } S_{pq} = 1, \quad (111)$$

with zeros outside the pattern.

ASDEX instead star-colors the symmetric Hessian graph. For each pair $\{p, q\}$, a hub-spoke orientation selects an uncontaminated compressed location and writes it to both H_{pq} and H_{qp} . This can reduce colors [27, 28]. Atom-block coloring can be lifted to coordinate colors $3c_i + d$, $d \in \{0, 1, 2\}$; availability depends on the installed ASDEX implementation.

The pipeline is summarized by

$$\mathbf{S} \xrightarrow{\text{color}} \mathbf{V} \xrightarrow{\text{HVPs}} \mathbf{Y} = \mathbf{H}\mathbf{V} \xrightarrow{\text{recover}} \mathbf{H}, \quad \text{ideal AD reduction} = \frac{3N}{n_c}. \quad (112)$$

Pattern setup, compilation, decompression, and GPU utilization limit the wall-time gain. Reuse a coloring only while its structural pattern remains valid.

The wrapper uses `atoms_to_inputs`, `sparsity_pattern`, `asdex.hessian_coloring_from_sparsity`, `get_energy_fn`, `get_hessian_fn`, and `jax.jit`. It densifies the sparse result and removes padding before eigendecomposition; only the AD stage exploits sparsity.

`no_shadow=True` stops adaptive-neighbor-selection derivatives and uses the selected graph. Shadow derivatives introduce extra couplings; `shadow=true` is rejected in this sparse path. Check consistency with MD forces. With three PET blocks, the paper derives $K = 7$, so the configured `hops=3` is an approximation requiring convergence.

Truncated ASD does not simply zero distant blocks of a dense Hessian: omitted couplings can share seeds with retained entries. For Cartesian mask $\mathbf{M}_k$ and reconstructed $\widetilde{\mathbf{H}}_k$ before acoustic treatment,

$$\Delta\mathbf{H}_{\text{disc}} = -(\mathbf{1} - \mathbf{M}_k) \odot \mathbf{H}, \quad (113)$$

$$\Delta\mathbf{H}_{\text{cont}} = \mathbf{M}_k \odot (\widetilde{\mathbf{H}}_k - \mathbf{H}), \quad (114)$$

$$\|\widetilde{\mathbf{H}}_k - \mathbf{H}\|_F^2 = \|\Delta\mathbf{H}_{\text{disc}}\|_F^2 + \|\Delta\mathbf{H}_{\text{cont}}\|_F^2. \quad (115)$$

Here $\odot$ is entrywise multiplication. The paper finds contamination tracking discarded norm down to roundoff; this is empirical, not a general bound. Acoustic-sum-rule enforcement does not remove either error.

At 0.1% heat-capacity tolerance at 300 K, most paper benchmarks converge at two hops for MACE, three for PET-S, and four for PET-XS, with slower zeolite cases [28]. Loaded MOF-5 requires its own comparison against exact-pattern or feasible dense AD, inspecting signed modes and the correction across temperatures. Convergence can be nonmonotonic near mode thresholds. Periodic image folding also requires cell/supercell convergence.

Table 5: Numerical controls for Hessian evaluation. Template settings are not established convergence values; archived results use float64.

Control	Effect	Template
<code>hops</code>	Changes the assumed sparsity approximation	3
<code>chunk_size</code>	Batches colored directions; affects memory/throughput	1
<code>remat</code>	Recomputes intermediates to reduce memory	true
<code>dtype</code>	Sets floating-point precision	float32

Keep separately tagged convergence outputs, including higher precision and less truncated references, especially for low-frequency modes.

4.5 Normal modes and quantum harmonic heat capacity

Symmetrize the Hessian, enforce the acoustic sum rule, and mass weight:

$$\sum_j H_{i\alpha,j\beta} = 0, \quad (116)$$

$$D_{i\alpha,j\beta} = \frac{H_{i\alpha,j\beta}}{\sqrt{m_i m_j}}, \quad \mathbf{D}\mathbf{e}_k = \omega_k^2 \mathbf{e}_k. \quad (117)$$

Uniform translations should have zero restoring force. Unlike the upstream SADMOF observable, which maps negative eigenvalues to zero, the local calculation preserves signed imaginary frequencies. Strict hybrid assembly rejects unexpected imaginary modes and allows at most three translational near-zero modes. Exploratory outputs instead discard modes at or below the threshold (Section 5.5).

For retained positive modes,

$$x_k = \frac{\hbar\omega_k}{k_B T} = \frac{h_{\text{PC}} \tilde{\nu}_k}{k_B T}, \quad (118)$$

$$C_{V,k} = k_B \left[\frac{x_k/2}{\sinh(x_k/2)} \right]^2, \quad (119)$$

$$c_V(T) = \frac{1}{M_{\text{cell}}} \sum_{k \in \mathcal{V}} C_{V,k}, \quad (120)$$

$\tilde{\nu}_k$ is wavenumber, and M_{cell} includes guests. Using cell mass in grams and mode heat capacity in J/K gives J/g/K. Equivalently use N_A/M_{molar} . One spectrum supplies the configured 100–500 K curve every 10 K; temperature changes populations, assuming fixed force constants.

4.6 Optimized minima and the hybrid correction

Relax the equilibrated empty structure directly at fixed cell for a separate reference. For loaded systems, quench representative independent replicas with the same MLIP and accept only force-converged, stable minima. Raw thermal frames are not harmonic references. Current outputs use one geometry per model/loading from 400 K.

For N_m accepted minima,

$$\overline{\Delta c^{\text{har}}}(T) = \frac{1}{N_m} \sum_{s=1}^{N_m} \left[c_{\text{qn},s}^{\text{har}}(T) - c_{\text{cl},s}^{\text{har}} \right], \quad (121)$$

$$s_{\Delta c}^2(T) = \frac{1}{N_m - 1} \sum_{s=1}^{N_m} \left[\Delta c_s^{\text{har}}(T) - \overline{\Delta c^{\text{har}}}(T) \right]^2. \quad (122)$$

Use the same retained modes for quantum and classical terms, with k_B per classical mode. Mixed host–guest motion requires the complete loaded spectrum.

$$c_P^{\text{approx}}(T) = c_{P,\text{cl}}^{\text{anh}}(T) + \overline{\Delta c^{\text{har}}}(T), \quad (123)$$

$$c_{P,\text{cl}}^{\text{anh}}(T) = \frac{1}{M_{\text{cell}}} \frac{d\langle H \rangle_{\text{cl}}}{dT}.$$

This combines an anharmonic classical C_P with a harmonic C_V correction, assuming a small quantum pressure–volume term. Validation below about 200 K remains limited.

Persistent LLPR members must contribute both terms before their spread is taken:

$$\sigma_{\text{model}}^2 = \sigma_{\text{cl,model}}^2 + \sigma_{\text{har,model}}^2 + 2 \text{cov}_i(c_{P,\text{cl}}^{(i)}, \Delta c_{\text{har}}^{(i)}), \quad (124)$$

$$u_{\text{hyb}}^2 = \sigma_{\text{MD}}^2 + \sigma_{\text{har,samp}}^2 + \sigma_{\text{model}}^2. \quad (125)$$

The covariance is across matched model members. σ_{MD} is the sampling error of the central enthalpy derivative; $\sigma_{\text{har,samp}}$ comes from independently quenched structures. Quadrature assumes independence of the sampling and model contributions.

Current single-replica analysis perturbs enthalpy means with Gaussian draws using autocorrelation-based errors and recomputes the derivative. It supplies within-trajectory error, not a block bootstrap or between-replica variance. Bands omit the unestimated harmonic sampling term: $u_{\text{hyb}} = (\sigma_{\text{MD}}^2 + \sigma_{\text{model}}^2)^{1/2}$. LLPR Hessians use central-model geometries without member-specific relaxation.

5 Results and discussion

5.1 Available calculations and status

The campaign contains 66 trajectories: PET-MAD (`pet-mad-1.5-s-40nn`) and PET-SOL (`pet-sol-s-best`), three loadings, eleven temperatures, and one replica per state point. Each 500 ps run contributes 8001 frames after discarding 100 ps: 33 ns simulated and 26.4 ns analyzed in total. Hybrid curves span 200–400 K; 175/425 K support centered endpoint derivatives. Each curve reuses one loaded 400 K reference spectrum.

All estimates are *exploratory*: every loaded spectrum has imaginary modes, and four of six relaxations failed their convergence criterion. Discarding unstable modes does not satisfy the strict acceptance protocol. Bands quantify selected sampling and model errors, excluding unstable-mode bias and unestimated variation across minima.

Trajectory diagnostics come from per-run `summary.json`; central values and relaxation diagnostics come from `heat-capacity.exploratory-discard-imaginary` CSV/JSON archives. Reported bands use refreshed stride-1 evaluation of all 8001 frames with 64 persistent LLPR members, rather than older per-loading uncertainty columns. Figure sources are recorded in `results-provenance.json`.

5.2 Thermodynamic sampling and equilibration

Mean temperatures stay within 0.315 K of target. PET-MAD/100 at 300 K gives 299.854 ± 8.031 K (mean and standard deviation), close to the equipartition width $T\sqrt{2/(3N-3)} = 8.058$ K. This checks kinetic sampling, not configurational ergodicity. Pressure fluctuations span $0.80\text{--}1.51 \times 10^3$ bar with mean errors of 9–18 bar; PET-MAD/100 gives 4.65 ± 13.11 bar, compatible with the 1 bar target. Physical enthalpy always uses imposed pressure.

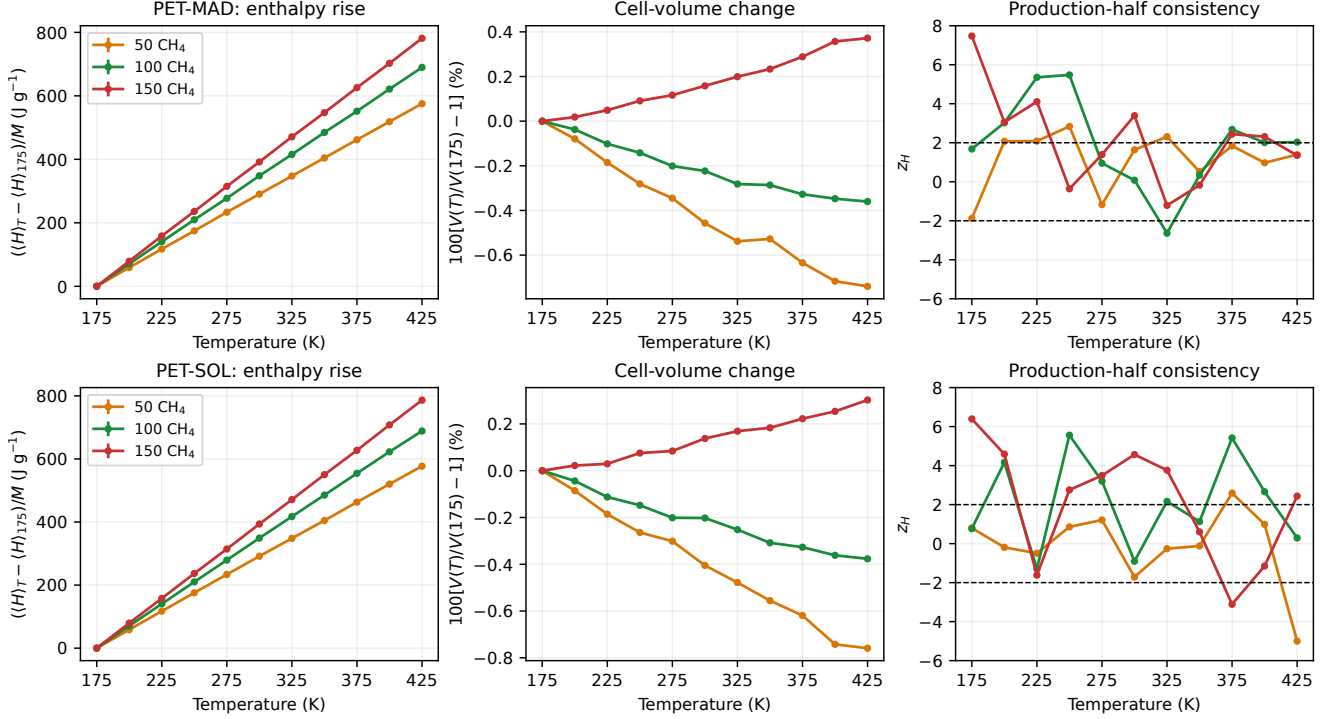


Figure 7: Thermodynamic diagnostics from all 66 trajectory summaries. Left: mass-specific enthalpy increase relative to the same model and loading at 175 K; the baseline is a display offset, and the small bars show the pointwise enthalpy standard errors without propagating the shared baseline error. Middle: fractional change of mean cell volume relative to 175 K. Right: the archived difference between production-half mean enthalpies divided by their combined standard error, z_H ; dashed lines mark ± 2 . These are diagnostics of the individual runs, not replica averages.

Enthalpy increases smoothly (Figure 7). Archived enthalpy correlation times of 0.095–0.101 ps imply 1980–2115 effective samples and mean errors of 0.015–0.046 eV. These short-time estimates do not exclude slow drift.

Half-trajectory discrepancies flag 33 of 66 runs with $|z_H| > 2$, where

$$z_H = \frac{\bar{H}_{\text{second}} - \bar{H}_{\text{first}}}{\sqrt{\text{SE}_{\text{first}}^2 + \text{SE}_{\text{second}}^2}}. \quad (126)$$

Examples are $z_H = 7.47$ for PET-MAD/150 at 175 K and 4.57 for PET-SOL/150 at 300 K. Some RMS-force correlation times reach 26.1 ps, leaving fewer than eight effective samples.

In Figure 8, the accumulated mean differs from the final mean by about -0.30 eV after 100 ps and -0.15 eV after 200 ps. A large model spread does not absorb a temperature-dependent sampling bias. Longer independent runs and sensitivity to discarded intervals are needed to validate the reported small MD errors.

5.3 Cell response, framework motion, and methane mobility

Both models predict contraction at 50/100 guests and expansion at 150 over 175–425 K (Figure 7). PET-MAD volumes change from 17146.8 to 17019.9, 17170.2 to 17108.4, and 17210.4 to 17274.4 $\text{\AA}^3$, respectively. Across models the changes are about -0.75% , -0.37% , and $+0.3\text{--}0.4\%$. Densities at 300 K are about 0.677, 0.752, and 0.825 g/cm³, largely reflecting added guest mass. Because $C_P - C_V \propto \alpha_V^2$, contraction does not imply a negative heat-capacity difference; compressibility is also needed to quantify it.

At 300 K, framework RMSDs are 0.659–0.725 $\text{\AA}$, while tracked bond-length changes are only 0.055–0.059 $\text{\AA}$. Bounded RMSD supports collective motion without gross collapse over the analyzed interval, but tracking a reference neighbor set cannot rule out all bond changes.

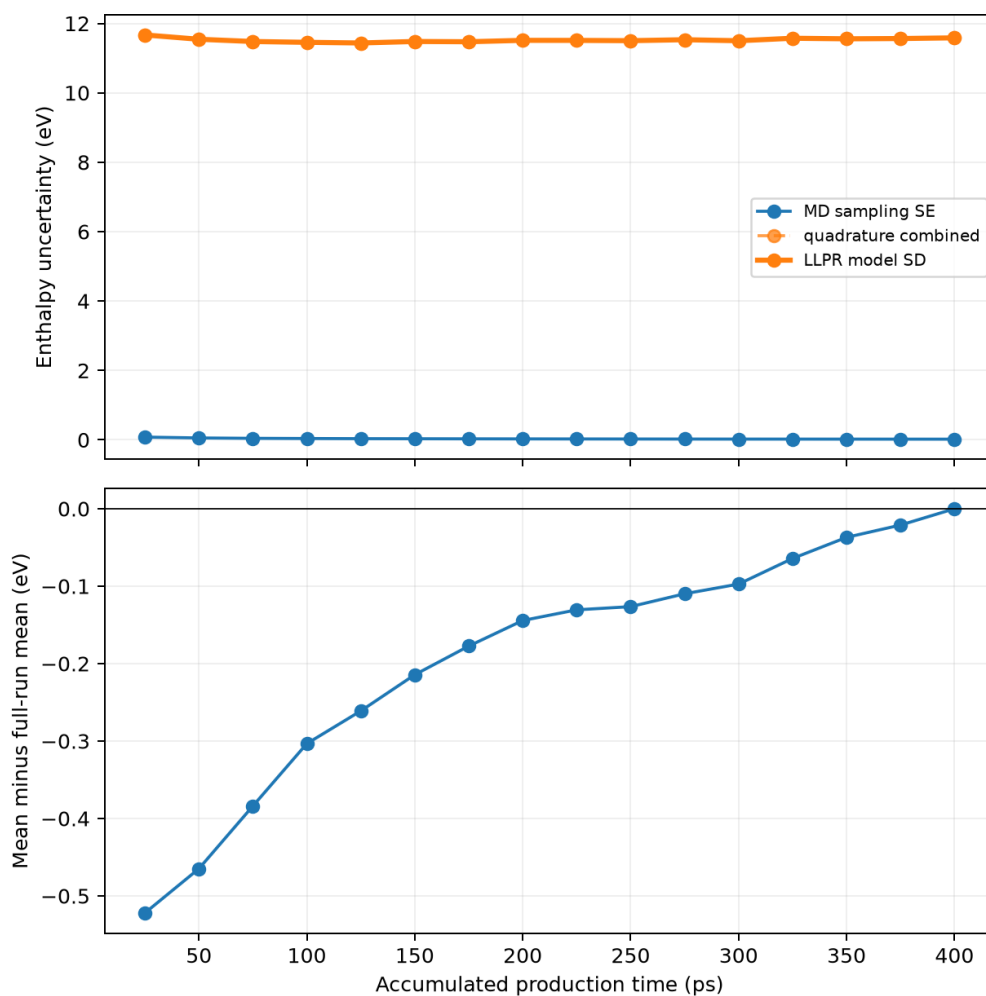


Figure 8: Archived enthalpy convergence diagnostic for PET-MAD with 150 methane molecules at 175 K. The upper panel separates MD sampling error from the LLPR spread of the *enthalpy*, in eV. The lower panel shows the accumulated mean relative to the full-production mean. Its endpoint is zero by construction and is not evidence of equilibration.

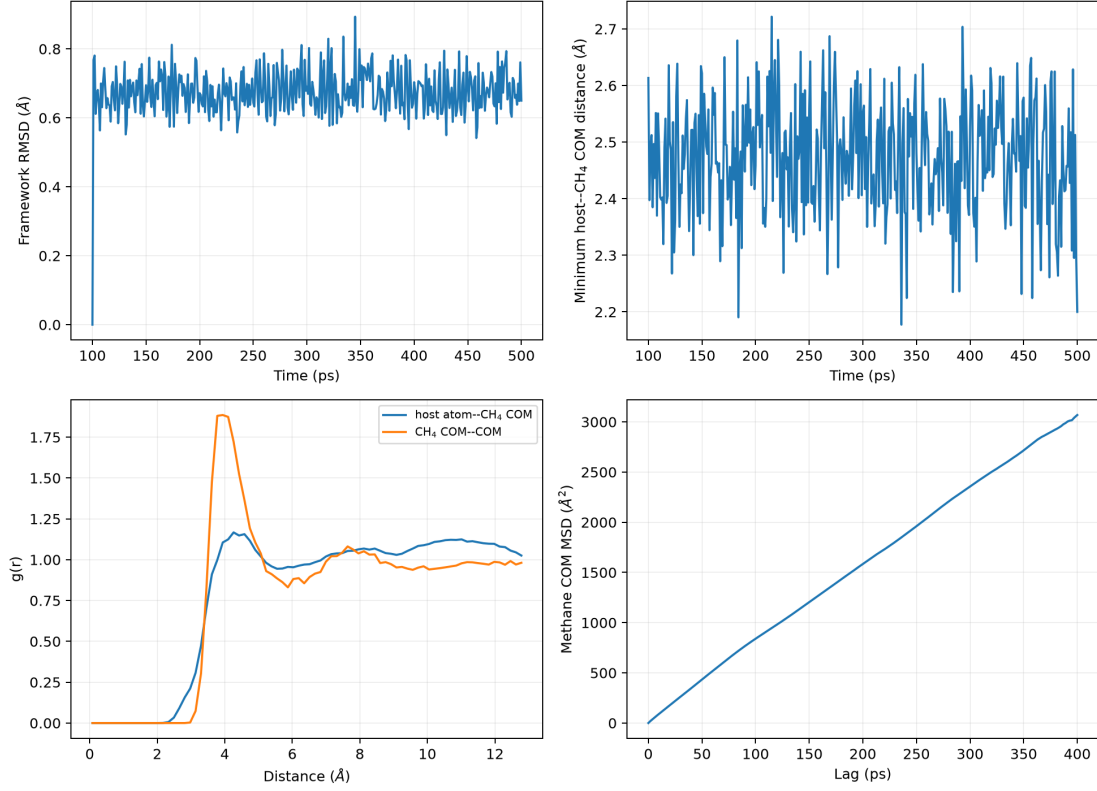


Figure 9: Structural diagnostics for PET-SOL with 150 methane molecules at 300 K: framework RMSD, nearest host-atom-methane center-of-mass distance, radial distributions, and methane center-of-mass mean-squared displacement (MSD). The first analyzed frame at 100 ps defines the framework reference, hence the initial zero RMSD. Long-lag MSD values use fewer time origins than short-lag values.

Methane 400 ps MSDs at 300 K decrease with loading: $8.9\text{--}9.2$, $5.1\text{--}5.6$, and $3.1\text{--}3.4 \times 10^3 \text{ Å}^2$ for 50, 100, and 150 molecules, respectively. Guest radial distributions peak near 4 Å . Mobility and crowding support the need for anharmonic MD; a single harmonic geometry cannot describe diffusion. Endpoint MSDs alone do not establish diffusion coefficients without a verified diffusive regime, cell-motion control, and replica statistics.

5.4 Classical, harmonic, and hybrid heat capacities

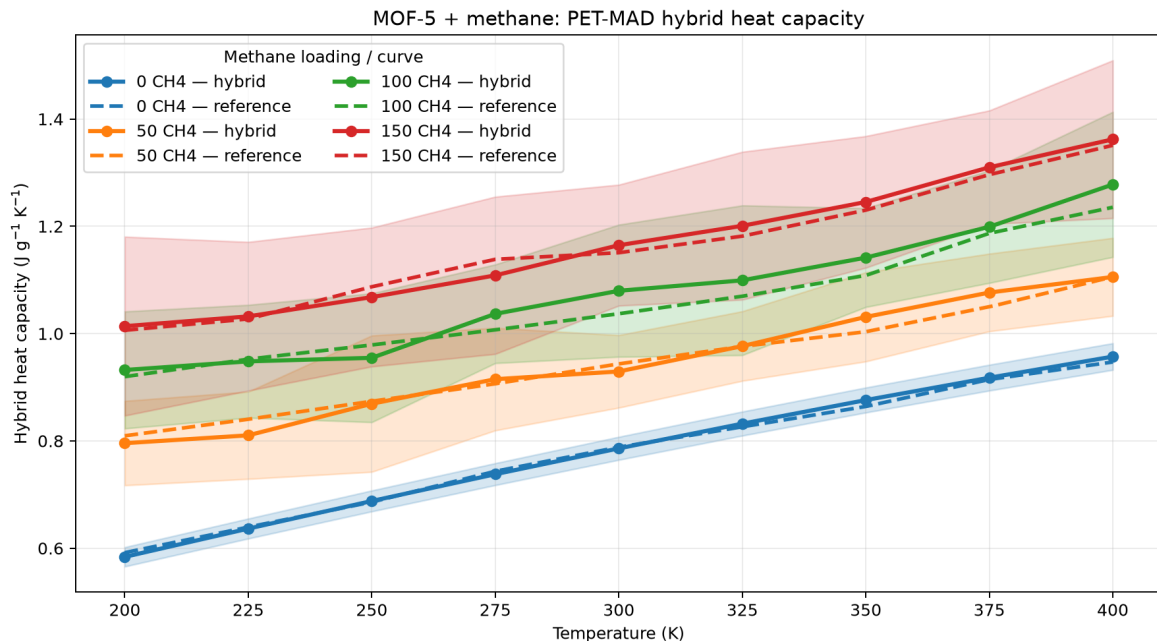
Table 6 resolves the 300 K contributions. The negative harmonic correction removes 59–63% of the classical term, as quantum occupations suppress high-frequency vibrations. Its increase in magnitude with loading reflects added methane and mixed host-guest modes, requiring loaded rather than empty spectra.

Table 6: Heat capacities and uncertainty estimates at 300 K, all in J/g/K and normalized by the total mass of each loaded cell. c_{cl} is the central enthalpy derivative, Δc_{har} the retained-mode harmonic correction, and c_{hyb} their sum. u_{CEA} and u_{G} use different paired model spreads around the *same* central value and MD standard error σ_{MD} . The final column is the separate variance-based central hybrid estimate. None of these uncertainties includes the bias from discarded imaginary modes or an estimated spread across minima.

Model	N_{CH_4}	c_{cl}	Δc_{har}	c_{hyb}	σ_{MD}	u_{CEA}	u_{G}	$c_{\text{hyb}}^{\text{var}}$
PET-MAD	50	2.281	-1.351	0.929	0.010	0.068	0.127	0.926
PET-MAD	100	2.762	-1.682	1.080	0.010	0.123	0.169	1.044
PET-MAD	150	3.122	-1.957	1.165	0.010	0.113	0.214	1.202
PET-SOL	50	2.289	-1.363	0.926	0.010	0.087	0.150	0.940
PET-SOL	100	2.762	-1.693	1.069	0.010	0.153	0.225	1.024
PET-SOL	150	3.134	-1.969	1.166	0.010	0.103	0.306	1.230

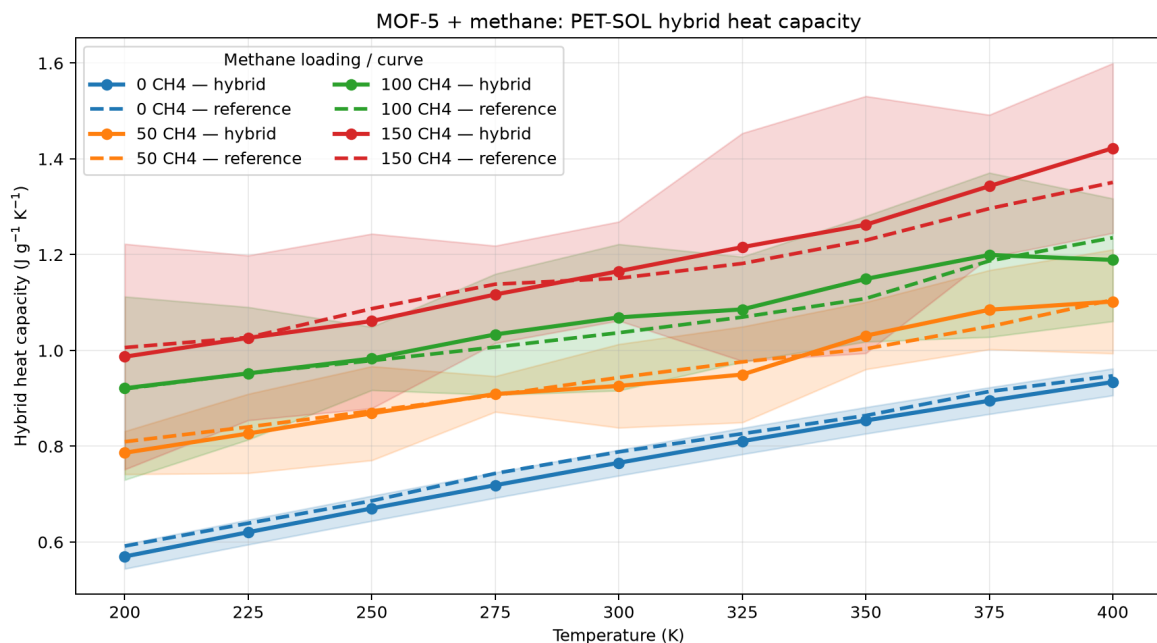
Room-temperature hybrid values rise from approximately 0.93 to 1.07–1.08 and 1.16 J/g/K with loading. Model differences of 0.003, 0.011, and 0.001 J/g/K are below propagated bands and cannot distinguish predictive accuracy.

The curves generally rise over 200–400 K (Figures 10–11). For PET-MAD/50, the classical term decreases from 2.347 to 2.269 J/g/K, while the correction becomes less negative, from -1.551 to $-1.164 \text{ J g}^{-1} \text{ K}^{-1}$. Thus the upward hybrid slope need not indicate increasing classical anharmonicity. Adjacent loading bands overlap;



Exploratory: unstable/near-zero modes discarded; loaded uncertainty includes sampling and LLPR spread.

Figure 10: Supplied PET-MAD loading comparison. Loaded solid curves are central classical enthalpy derivatives plus retained-mode harmonic quantum corrections. Their shading is $\pm u_{\text{CEA}}$, combining MD sampling error with correlated classical-harmonic LLPR spread. Despite the common “hybrid” legend, the zero-loading curve is a quantum-harmonic c_V calculation only, with a harmonic LLPR band; no empty-system MD term was computed. Dashed curves are the supplied reference CSVs, whose provenance is discussed in the text.



Exploratory: unstable/near-zero modes discarded; loaded uncertainty includes sampling and LLPR spread.

Figure 11: Supplied PET-SOL loading comparison, with the same definitions and exploratory mode treatment as Figure 10. The slight decrease of the 100-molecule central curve at the upper end of the grid is smaller than its uncertainty band and is not evidence of a resolved thermodynamic anomaly. Shading denotes standard uncertainty, not a calibrated confidence interval.

significance requires paired propagation of their difference. The grid does not bracket the reference low-temperature minimum, and small local bends remain unresolved within uncertainty.

Empty-framework 300 K values are 0.786 ± 0.021 (PET-MAD) and 0.766 ± 0.027 J g⁻¹ K⁻¹ (PET-SOL). These are framework-mass-normalized quantum harmonic c_V values with LLPR spread, not hybrid c_P or incremental adsorption heat capacities.

Dashed curves are digitized Kapil *et al.* data [16]. Their 300 K values are approximately 0.788, 0.944, 1.037, and 1.151 J/g/K for 0/50/100/150 guests. Loaded differences from this work are at most 0.043 J/g/K. Agreement is contextual: the digitization has no uncertainty estimates, the force fields and preparation differ, and present sampling/spectra remain unconverged. The earlier approximate empty-host scale is not a separate fitting target.

5.5 Hessian stability

Table 7: Archived central loaded-Hessian diagnostics. All calculations use PET-JAX, float64, three graph hops, and the force-constant row-sum acoustic treatment. “Converged” reproduces the relaxation archive’s flag for its requested tolerance. Imaginary modes have signed frequencies below -1 cm⁻¹; near-zero modes satisfy $|\tilde{\nu}| \leq 1$ cm⁻¹. Retained modes have positive frequencies above the threshold. Signed negative values represent unstable curvature, not physical negative-frequency oscillators.

Model	N_{CH_4}	Converged	F_{max} (eV/Å)	Imaginary	Near zero	Retained	$\tilde{\nu}_{\text{min}}$ (cm ⁻¹)
PET-MAD	50	no	0.0170	17	5	2000	-53.5
PET-MAD	100	yes	0.0097	114	3	2655	-92.4
PET-MAD	150	no	0.0095	28	3	3491	-66.8
PET-SOL	50	no	0.0054	32	3	1987	-107.7
PET-SOL	100	yes	0.0100	95	3	2674	-138.4
PET-SOL	150	no	0.0049	35	3	3484	-86.5

Force convergence does not establish positive curvature: the two 100-molecule structures pass their archived force criteria but retain 114/95 imaginary modes, down to $-92.4/ -138.4$ cm⁻¹. Other loaded runs have 17–35 imaginary modes and failed convergence flags; PET-MAD/50 also has five near-zero modes. Run-dependent criteria must be judged from each archive rather than comparing residual forces alone. Empty spectra have four/five imaginary modes, down to $-5.5/ -6.6$ cm⁻¹.

Discarding negative eigenvalues enables an oscillator sum but does not create a stable minimum. Distinguish insufficient relaxation, unstable stationary structures, and sparse-curvature errors with convergence checks. Matching quantum/classical mode masks avoids inconsistent subtraction; it does not estimate unstable-mode bias. Reusing one 400 K geometry also leaves temperature dependence and minimum-to-minimum variation untested. LLPR spread at a central geometry does not replace structural sampling.

5.6 Uncertainty bands

Figures 12–13 separate MD sampling, classical LLPR, harmonic LLPR, and paired total uncertainty. MD errors are about 0.010 J/g/K at 300 K; model terms dominate. Covariance can make the total smaller than an individual component (Eq. (124)).

For PET-SOL with 150 guests at 300 K, classical and harmonic CEA spreads of 0.108 and 0.155 J/g/K combine to 0.103 J/g/K, with inferred correlation -0.75 . PET-SOL/100 instead has correlation $+0.37$ and paired spread 0.153 J/g/K. Independent quadrature would miss both effects; predictive spread is not reduced by $\sqrt{64}$.

Gaussian propagation gives larger 300 K bands in every case (Table 6), with positive classical–harmonic correlations of 0.43–0.63. Smoother or narrower bands do not demonstrate calibration. Recorded $\text{var}(\beta\Delta U_i)$ spans about 2.4 to 1.42×10^3 , violating the small-variance CEA criterion; some direct reweighting effective counts approach one. Neither collapsed reweighting nor untested Gaussian tails validate these bands, whose coverage against reference electronic-structure data remains unestablished.

The separate **variance-hybrid** calculation also changes the central classical estimator. At 300 K, hybrid differences from the derivative route range from -0.045 to $+0.064$ J g⁻¹ K⁻¹, with MD errors of 0.022–0.030 J/g/K from squared-fluctuation correlations. Shared trajectories and harmonic corrections make these cross-checks correlated. Quantitative predictions require stable Hessians, stationary independent trajectories, multiple quenched structures, and validated member propagation.

6 Recommended production and analysis protocol

Validate timestep through short NVE energy-drift tests before reservoir coupling, which can mask integration errors. Cell relaxation requires a stress-capable, validated model. One-cell Hessians and selected minima need

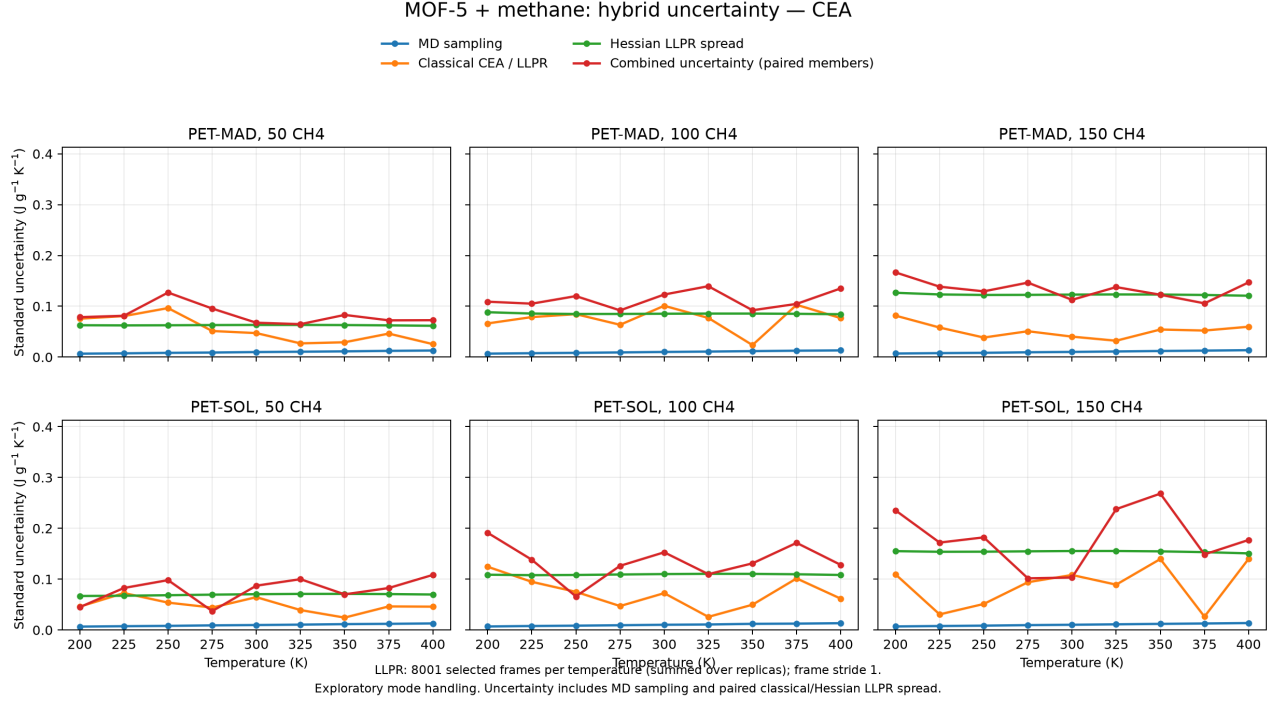


Figure 12: Supplied uncertainty decomposition using refreshed CEA member curves. Every panel uses all 8001 saved production frames at each temperature and 64 LLPR members. The harmonic curve is model spread at one fixed geometry, not an uncertainty across minima. The total preserves classical-harmonic covariance and excludes an unestimated harmonic sampling term. All plotted quantities are standard errors or model standard deviations in J/g/K, not variance fractions.

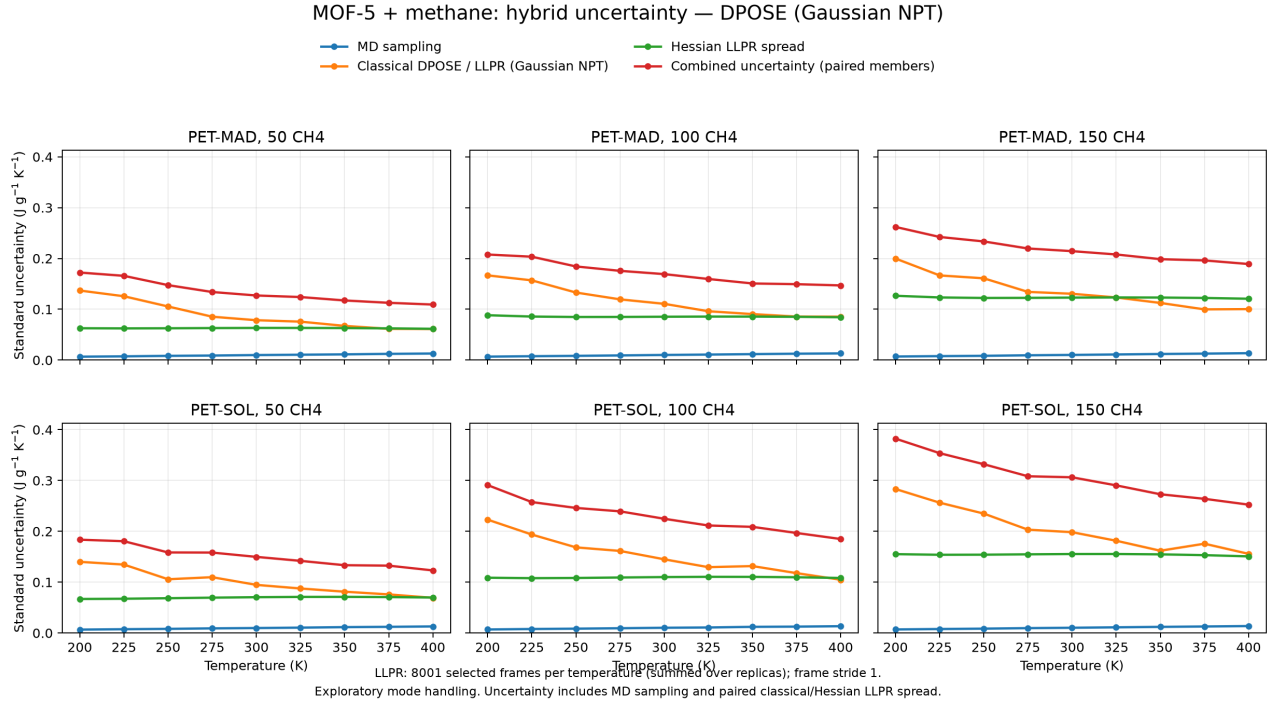


Figure 13: Supplied Gaussian NPT/DPOSE uncertainty decomposition. Central hybrid curves, MD sampling errors, harmonic member corrections, and frame coverage are identical to the CEA comparison; only the classical member spread and its covariance with the harmonic correction change. The smoother orange curves result from a different estimator and do not establish more accurate uncertainty calibration.

Table 8: Production requirements for the workflow in Figure 1. Reference agreement supplements internal convergence; it is not a fitting target.

Calculation	Acceptance and convergence requirements
Empty reference	Relax the equilibrated host directly at fixed cell with the selected checkpoint. Report forces and signed modes; retain a separately normalized harmonic curve.
Loaded NPT	Independent placement/velocity seeds. Check stationary enthalpy, cell, bonds, pores, and guest motion; converge duration, replicas, cell size, timestep, coupling times, and temperature spacing.
Loaded minima	Quench representative independent configurations at fixed cell. Reject unconverged or unstable structures.
Hessians	Converge hops, precision, HVP batching, acoustic treatment, frequency threshold, and supercell/phonon sampling on the correction.
Hybrid assembly	Average accepted loaded corrections; differentiate replica enthalpies; pair LLPR indices. Report MD, minimum sampling, and model uncertainties separately and jointly.

finite-size and configurational convergence; their unweighted spread is not automatically a free-energy-weighted average. Transfer of the reference hybrid approximation to PET, especially below 200 K, needs validation.

7 Implementation, outputs, and reproducibility

`mof_heat_capacity.protocols.loaded` prepares campaigns; `scripts/md/submit_loaded_md.sh` submits loaded NPT and rejects zero-methane MD. Property stages use the scripts under `scripts/properties/`: `submit_analysis.sh` for diagnostics, `submit_heat_capacity.sh` for fixed-cell relaxation and PET-JAX Hessians, and `submit_hybrid_analysis.sh` for final assembly. Hessian LLPR members run as eight GPU array tasks of eight members each; `analysis.llpr_merge` supplies the final hybrid dependency.

Archives retain structures, signed frequencies, model/run identities, grids, precision, hops, chunking, and re-materialization. Hybrid NPZ/CSV/JSON outputs store the classical term, harmonic correction, sum, uncertainties, and provenance.

Table 9: Uncertainty components, their calculation locations, and their stored outputs.

Quantity	Where it is calculated	Stored result
LLPR energy standard deviation	metatrain LLPR covariance and calibration	<code>energy_uncertainty</code> ; copied as <code>analytical_energy_uncertainty_eV</code> in the per-run archive
CEA member enthalpies and overlap diagnostics	<code>analysis/uncertainty.py</code> , function <code>committee_enthalpy_estimates</code>	per-run <code>model_uncertainty.npz</code>
Classical LLPR C_P standard deviation	<code>analysis/results.py</code> , after member-wise replica averaging and temperature differentiation	<code>model_uncertainty_heat_capacity.npz</code> and <code>.csv</code>
MD sampling standard error	<code>analysis/statistics.py</code> , then replica combination and derivative resampling in <code>analysis/hybrid.py</code>	<code>classical_anharmonic_cp_standard_error</code>
Harmonic sampling standard error	<code>analysis/hybrid.py</code> , across independent optimized minima	<code>harmonic_quantum_correction_standard_error</code>
Hessian LLPR standard deviation	<code>analysis/harmonic.py</code> , after differentiating persistent LLPR readouts at the central minimum	<code>harmonic_quantum_correction_model_standard_deviation</code>
Combined standard uncertainty	paired model variance and sampling quadrature in <code>analysis/hybrid.py</code>	<code>approximate_cp_combined_standard_uncertainty</code>

The scalar energy uncertainty is diagnostic; property errors require complete persistent member curves. Paired MD/harmonic readouts preserve covariance, but Hessians use central-model minima without member-specific relaxation.

Use one node, one task, and one GPU per MD/Hessian job; parallelize independent states, replicas, minima, or member tasks. Retain Slurm IDs and measure representative runtime/memory before adjusting requests:

```
sacct -j JOB_ID --format=JobID,State,Elapsed,MaxRSS,AllocTRES
```

Use distinct outputs for every loading, temperature, seed, and replica.

Record structure/composition and insertion criteria; checkpoint, export, conversion, and environment identities; LLPR dataset/ensemble hashes, calibration, regularizer, seeds, member count, stride, and overlap diagnostics;

ensemble/integration settings and restart history; relaxation forces, signed modes, and Hessian controls; and production intervals, replica counts, correlation/block methods, and convergence comparisons.

8 Limitations and interpretation

Table 10: Limits of the present estimates and the evidence needed to resolve them.

Present limitation	Required evidence
33/66 enthalpy drift flags; one replica per state	Longer independent trajectories and stable block/replica statistics.
Imaginary modes in every loaded reference; four failed relaxations	Stable minima and converged signed spectra, sparsity reach, and precision.
One loaded geometry per curve	Multiple independent quenches and sensitivity to harmonic-reference temperature.
Classical MD plus harmonic C_V correction	Validation of quantum anharmonic and pressure-volume effects, especially below 200 K.
Fixed-loading random insertion	Demonstrated equilibration of guest distributions; separate particle-exchange sampling for adsorption equilibrium.
Shallow LLPR; large CEA diagnostics and method-dependent bands	Reference-data calibration and validated propagation; shared representation bias and member-specific relaxation remain outside the current spread.
PET differs from QuickFF; finite-cell/numerical approximations	Loaded-MOF energy/force/stress/curvature benchmarks and size/grid convergence.

9 Conclusions

PET-MAD and PET-SOL give exploratory room-temperature heat capacities rising from about 0.93 to 1.16 J/g/K with methane loading. Harmonic quantization removes 59–63% of the classical term; guest mobility supports retaining anharmonic MD. Broad agreement with digitized reference trends does not establish quantitative accuracy. Stable spectra, independent trajectory/minimum sampling, and validated paired uncertainty propagation are needed before interpreting these curves as converged predictions.

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